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Internal information quality and performance metric selection

Wayne Guay[a,*,1], Chongho Kim[b], Oscar Timmermans[c]

[a] The Wharton School, University of Pennsylvania, United States
[b] Seoul National University, South Korea
[c] London School of Economics, United Kingdom

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We examine the role of firms' internal information quality (IIQ) in designing executive incentive contracts. We find that higher IIQ is associated with a greater number of performance metrics and increased dissimilarity from peer firms' contracts, particularly along non-financial dimensions. These relations hold when we examine changes in IIQ that are likely induced by plausibly exogenous shifts in two financial accounting standards. We further find that incorporating more numerous and more dissimilar non-financial metrics is positively associated with future profitability, but only when IIQ is high. Our results are consistent with the hypothesis that the quality of a firm's internal information is a friction in performance metric selection.

1. Introduction

The design of executive incentive-compensation contracts is one of the most important elements of effective corporate governance. A key feature of these contracts is the choice of performance metrics against which executives will be explicitly evaluated and paid. For many years, the literature largely emphasized stock-price-based performance metrics given that the lion's share of executives' incentives was shown to be tied to stock price via restricted stock and option grants and vested equity holdings.[2] However, the importance of non-price performance metrics has seen a resurgence—in part due to the increasing use of performance-vested equity awards, which surged after the adoption of ASC 718 in 2006, and in part due to the recent use of non-financial performance metrics in bonus plans.

Given the increasing importance of non-price performance metrics, it is important to understand the determinants of boards'

* Corresponding author.
E-mail addresses: guay@wharton.upenn.edu (W. Guay), chongho.kim@snu.ac.kr (C. Kim), o.timmermans@lse.ac.uk (O. Timmermans).

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2 See, for instance, Jensen and Murphy (1990b, 1990a), Hall and Liebman (1998), Core and Guay (1999), Bushman and Smith (2001), Core et al. (2003), Coles et al. (2006), and Edmans et al. (2017).

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selection of such metrics. Although no doubt a number of forces are at play, we focus on the role of internal information quality (IIQ), which may affect boards' ability to identify, measure, and track performance metrics in executive incentive contracts. Motivation for studying IIQ comes from the observation that, since the 2006 introduction of the Compensation Discussion and Analysis (CD&A) in public firms' proxy statements, the selection of performance metrics has taken on an important disclosure role by providing information to investors.[3] For this relatively underexplored role of performance metric choice as a disclosure mechanism, IIQ is a natural determinant to examine given prior findings that IIQ forms the foundation of external disclosure quality in various contexts.

Furthermore, IIQ is crucial for understanding the practical application of fundamental agency theory underlying executive incentive-compensation when information processing constraints exist. With regard to selecting performance metrics, prior studies generally assume that all firms can choose from the same opportunity set of performance metrics when designing incentive contracts for their executives. For instance, when examining whether and why firms include pro-forma earnings metrics or environmental and social performance metrics in their executive incentive contracts, practitioners and researchers often implicitly assume that the cost of including such metrics is the same across firms.[4] However, many performance metrics require metric-specific information processing, and non-financial performance metrics may lack the level of long-standing understanding among executives, directors, and investors that financial performance metrics enjoy. Reflecting this heterogeneity, we observe considerable differences across firms—even within the same industry—in both the number and variety of performance metrics included in executive incentive contracts.

In this study, we conjecture that the quality of a firm's internal information system is a limiting friction that constrains the breadth and variety of explicit performance metrics used in executive incentive contracts. That is, although economic theory predicts that any performance metric that provides additional information about the executive's actions is valuable in a frictionless world (Holmström, 1979), in practice there are processing costs for each candidate metric in the contract. Similar to the information processing constraints faced by investors (e.g., Blankespoor et al., 2020), these costs relate to, for instance, defining the metric, determining performance targets, collecting data to calculate the metric, and assessing and reacting to performance relative to the targets. Given these considerations, we hypothesize that improvements in firms' IIQ reduce processing costs for performance metric selection, enabling boards to use a broader set of performance metrics in their executives' incentive contracts.[5]

To study the relation between IIQ and performance metric selection, we focus on an IIQ index that combines five well-established measures from prior literature with two additional measures that capture aspects of firms' non-financial information environment. Specifically, our index incorporates variation related to the accuracy of executives' forecasts, the speed at which executives release the earnings announcement after fiscal year closing, audit fees, the absence of Sarbanes-Oxley Section 404 material weaknesses, the absence of restatements, environmental initiatives, and ESG reporting scope. This composite measure is intended to capture the quality of firms' overall information acquisition processes, internal control practices, and the overall efficacy of information systems—in which both financial and non-financial aspects play a complementary role.[6]

We begin our empirical analysis by examining whether firms' IIQ is positively correlated with the number of performance metrics incorporated into CEO incentive contracts. We show that this correlation exists and continues to hold after controlling for a comprehensive set of time-varying executive, firm, and board characteristics, alongside firm and year fixed effects. In addition, we find that IIQ is positively associated with how dissimilar CEO incentive contracts are compared to those of industry peers, suggesting that IIQ helps firms develop a firm-specific structure for their incentive contracts. Overall, these findings provide descriptive evidence suggesting that higher IIQ enables boards to increase the breadth of their performance metrics, which theory predicts allows them to more effectively address the specific agency conflicts their firms face.

Next, we conduct cross-sectional analyses to identify circumstances in which IIQ is more likely to function as a binding friction that limits the breadth and variety of explicit performance metrics included in CEO incentive contracts. We first identify subsamples of firms where we expect greater need for a broader and more tailored set of performance metrics and then assess whether IIQ plays a differential role in these settings. We find that the relation is strongest among firms: (1) characterized by greater organizational complexity; (2) for which internal information is easier to process and act upon; and (3) facing heightened external environmental and social pressures. These patterns suggest that limited IIQ constrains boards' ability to identify and implement relevant performance

[3] For instance, studies find that boards consider investor reactions when designing executive contracts and writing the CD&A (e.g., Laksmana et al., 2012; Gipper, 2021; Packard et al., 2023). The CD&A was introduced as part of revised executive compensation disclosure requirements issued by the U.S. Securities and Exchange Commission (2006).

[4] Regarding practitioners, Institutional Shareholder Services Inc (2024) considers the absence of both long-term and short-term performance metrics and incentives related to ESG criteria as a significant concern in executive incentive contracts. Relatedly, Amundi (2023), the largest asset manager in Europe, has voted against the remuneration of Adobe Inc. and other firms, due to the lack of ESG KPIs in the executive incentive contracts. Regarding researchers, studies argue that linking executive compensation to environmental and social performance metrics can improve firms' environmental and social outcomes and align managerial incentives with stakeholder interests, while largely focusing on the incentive effects of such metrics rather than potential constraints that may limit boards' ability to design and implement these types of performance metrics (e.g., Cordeiro and Sarkis, 2008; Hong et al., 2016; Flammer et al., 2019).

[5] These arguments are somewhat related to the long-standing theoretical and empirical literature on the effects of "signal" and "noise" on the relative weights placed on accounting earnings and stock prices in compensation contracts. In this study, our focus is not on a horse race between performance metrics with respect to which metric better captures executives' actions, but rather on understanding whether the quality of firms' internal information environment allows firms to use a broader set of performance metrics.

[6] See, for instance, Cassar and Gibson (2008), Feng et al. (2009), Plumlee and Yohn (2010), and Dorantes et al. (2013) for previous work that uses these measures to proxy for the quality of firms' financial and Simons (1990), Kaplan and Norton (1992), and Chenhall (2003) for previous work that examines the complementary role between financial and non-financial aspects of information systems.

metrics precisely when the demand for customization is greatest or when it is easier to process the available information—highlighting its role as a friction in performance metric selection.

Of course, firms' performance metric selection and their internal information environments are endogenously determined and therefore our main and cross-sectional results mainly capture associations. To supplement these findings, we next examine how shocks to firms' IIQ—induced by shifts in financial accounting standards (specifically ASC 606 "Revenue from Contracts with Customers" and ASC 842 "Leases")—affect performance metric selection. The goal of this alternative design, also used by prior literature (e.g., Gallemore and Labro, 2015; Shroff, 2017; Cheng et al., 2018; Heitzman and Huang, 2019), is to leverage plausibly exogenous variation in IIQ that enhances the credibility of our empirical design.[7] Consistent with previous work, we first show that these accounting standard shifts result in improvements in firms' IIQ. We next find that these plausibly exogenous shocks to IIQ coincide with an increase in the number of performance metrics in CEO incentive contracts, as well as a greater divergence from the performance metrics used by industry peers—particularly along financial accounting-related dimensions. We interpret these findings as evidence that when firms upgrade their financial reporting systems in preparation for the new accounting standards, they also learn about their operations and value chains and/or identify new financial performance dimensions that can be incorporated into their executive incentive contracts.

In two final analyses, we examine: (1) whether the role of IIQ differs between financial and non-financial performance metrics; and (2) whether previously documented negative performance effects of using a broad set of performance metrics are mitigated when firms have higher IIQ. Regarding the first analysis, we find that our main specification has a stronger association with non-financial metrics. This finding is consistent with the idea that non-financial metrics are less understood and harder to process than financial metrics, on average. Regarding the second analysis, we supplement prior findings (Albuquerque et al., 2025) by documenting that a broader and more dissimilar set of metrics—particularly non-financial metrics—is positively associated with future profitability, but only when used in conjunction with high IIQ. These findings highlight that the relations between the breadth and variety of performance metrics in CEO incentive contracts and future performance are most positive when accompanied by higher IIQ, which stands in contrast to other drivers of metric adoption such as pressure from compensation consultants and proxy advisors (Albuquerque et al., 2025).

Our study contributes to the large literature that examines how firms select performance metrics for their executive incentive contracts. Our study extends this literature by examining how information environment frictions contribute to the explicit form of executive incentive contracts. Such frictions are likely to be particularly salient given the recent surge in non-financial performance metrics such as environmental and social metrics, which typically lack the level of long-standing understanding financial performance metrics enjoy. Our study also supplements prior studies that examine drivers of executive incentive contracts that do not stem from their efficacy in dealing with agency conflicts, such as external pressure from compensation consultants and proxy advisors (Albuquerque et al., 2025; Cabezon, 2025), and provides additional insights into the role of the breadth and variety of performance metrics. More broadly, our study relates to research showing that accounting considerations influence the structure of executive incentive contracts (e.g., Carter et al., 2007; Cadman et al., 2013). Supplementing this literature, our quasi-experimental evidence shows how changes in financial accounting standards relate to performance metric selection through changes in IIQ.

Our study also contributes to understanding the explicit performance metrics used in performance-vested equity awards, which have become increasingly important since the adoption of ASC 718 in 2006 (Bettis et al., 2018; Pawliczek, 2021). Closer to our study, Core and Packard (2022) document that the spread of performance-vested equity awards strengthened agency theory's prediction that noisier metrics receive less weight in incentive contracts (Banker and Datar, 1989). In contrast, we focus on another agency theory prediction that any performance metric that can provide additional information about the agent's action should be included in incentive contracts absent frictions (Holmström, 1979) and provide new insights into the frictions behind the usage of performance-vested equity awards.

Finally, our study extends the literature on the measurement and disclosure implications of firms' internal information environment. With respect to measurement, we incorporate two non-financial dimensions into a composite measure and show that this broader proxy is associated with firms' use of non-financial performance metrics in executive incentive contracts. In doing so, we highlight that IIQ measures capture not only financial reporting efficacy but also the broader relevance of management practices. Furthermore, we extend the literature that examines how IIQ affects external disclosure by highlighting an underexplored nature of executive incentive-compensation design as a disclosure channel for investors.

The paper is organized as follows. Section 2 discusses related literature and develops hypotheses. Section 3 describes our data. Sections 4 through 6 present, respectively, our main, quasi-experimental, and additional tests and results. Section 7 concludes.

[7] Although these shocks offer plausibly exogenous variation in firms' internal information environments, we do not intend to convey that all changes to executive incentive contracts stem from exogenous shocks like these. Further, we fully acknowledge that the direction of causality between IIQ and performance metric selection is likely to run in both directions. That is, exogenous variation in IIQ may dictate the selection of performance metrics from the supply side as the marginal cost of using performance metrics decreases, but so too might exogenous variation in the demand for specific performance metrics dictate investments in IIQ. The latter demand-driven mechanism may still suggest that IIQ is a necessary condition for effectively using performance metrics, which is at least narrowly consistent with our interpretation of the role of IIQ for incentive contracts, but we fully acknowledge that there can be important nuances. We view both of these effects as interesting. We attempt to shed more light on the former in this study and mostly leave the latter to future research.

2. Related literature and theory development

2.1. Related literature

Since the inception of agency theory, the set of performance metrics included in executive incentive contracts and their optimal weights have been of great importance. At a fundamental level, Holmström (1979) shows that in a frictionless agency conflict environment in which adding more performance metrics to the contract is costless, any performance metric that provides additional information about the agent's action is valuable (i.e., results in a more efficient contract). Subsequent studies have added more frictions to this ideal setting, such as limited ways of aggregating accounting information (Banker and Datar, 1989) and lack of performance metrics' congruity in multi-action settings (Feltham and Xie, 1994) to examine how the optimal weights of performance metrics change with such frictions.

Initially, such interest in performance metrics was mainly based on their empirical use in cash compensation in incentive contracts. As cash compensation lost its significance because of the advent of time-vested equity awards, the importance of performance metrics also waned.[8] More importantly, although the weights of performance metrics in cash compensation were seemingly consistent with agency theory (i.e., noisier metrics have less weight), that pattern did not extend to total compensation (e.g., Core et al., 2003), reducing the importance of studying explicit performance metrics that are concentrated in cash compensation.

However, explicit performance metrics regained significance as their usage in performance-vested equity awards started to gain traction. Performance-vested equity awards became more important in executive compensation after the adoption of ASC 718 in 2006 that made accounting treatments of performance-vested and time-vested equity awards more equalized. Over time, the prevalence of performance-vested equity awards rose to be used by a majority of firms, making them responsible for roughly half of all equity awards.[9] The recent substantial increase in the number of performance metrics used in short-term and long-term performance plans is well evidenced and sparked the realization that performance metrics serve incentive, coordination, and signaling roles that were previously unexplored.

Combined with the growing interest in the use of non-financial (e.g., environmental and social) metrics in executive compensation, this trend has also attracted academic interest. For example, Core and Packard (2022) document that the spread of performance-vested equity awards strengthened the agency theory's prediction that noisier metrics receive less weight in incentive contracts, and Cohen et al. (2023) argue that firms use environmental and social metrics to credibly signal to investors their commitment to focus on these non-financial aspects.[10] Overall, this recent development in the literature suggests that examining explicit performance metrics again has the potential to provide empirical insights about the application of agency theory in executive compensation. Reinforcing this trend, recent studies suggest that cash compensation provides stronger incentives than previously considered or serves different purposes than equity-based compensation, further motivating the study of explicit performance metrics in executive pay design.[11]

2.2. Hypothesis development

Fundamental agency theory argues that the value of additional performance metrics in incentive contracts primarily comes from more efficient risk sharing between the principal and the agent (Holmström, 1979; Banker and Datar, 1989). As such, a performance metric is valuable when it provides information about the agent's action not captured by other performance metrics, without imposing excessive risk on the agent due to the metric's variation unrelated to that action. However, although theoretically sound, in practice each candidate metric in a contract incurs processing costs, such as defining the metric, setting performance targets, gathering data for calculation, and evaluating performance against targets.

Given these considerations, we hypothesize that improved IIQ reduces the processing costs associated with performance metrics. We predict that this reduction, in turn, allows boards to incorporate a broader set of performance metrics into executive incentive contracts. In addition, lower processing costs allow boards to design executive incentive contracts that better cater to firms' idiosyncratic situation and thus are dissimilar from those of industry peers, as boards can rely less on peer practices when selecting and implementing metrics that are otherwise difficult to process. Viewing the performance metrics in an incentive contract as an equilibrium between the demand for and supply of such metrics, this discussion mainly suggests that IIQ increases the supply side by reducing the marginal cost of using performance metrics.

This discussion leads to our central hypothesis (stated in alternative form) that improvements in IIQ make it more likely for boards to increase the breadth and variety of performance metrics in their design of CEO incentive contracts:

Hypothesis 1. Higher quality internal information environments are positively associated with the breadth and variety of performance metrics in a CEO's incentive contract.

[8] See, for instance, Bushman and Smith (2001), Core et al. (2003), and Armstrong et al. (2010).

[9] See, for instance, Bettis et al. (2018), Pawliczek (2021), and Core and Packard (2022).

[10] In addition, prior studies suggest that non-financial metrics may have been on the rise due to them enabling executives' rent extraction (Bebchuk and Tallarita, 2022), although there are also counterarguments (Cohen et al., 2023). However, this possibility of rent extraction is unlikely to be related to our study as higher IIQ is less likely to facilitate rent extraction through having more performance metrics.

[11] See, for instance, Guay et al. (2019), Bloomfield (2021), Bloomfield et al. (2021), Timmermans (2024), Armstrong et al. (2025a), and Armstrong et al. (2025b).

Although we expect every firm to benefit from exploiting their high quality internal information, we posit that these benefits differ across firms depending on the extent to which other frictions constrain boards' ability to tailor executive incentive contracts. For instance, economic theory predicts that these benefits depend crucially on boards' ability to process the information and the usefulness of having multiple performance metrics with respect to their informativeness and firms' circumstances. We examine three broad areas of these determinants: organizational complexity, the ease of internal information processing, and external pressures. These factors are conceptually distinct from those that directly influence the quality of firms' internal information environment. Rather, they determine boards' demand or need for more customized metrics: for instance, due to the complexity of operations, the interpretability of relevant information, or the external pressures to address stakeholder concerns.

The overarching argument across these dimensions is that when boards face little need to tailor incentive contracts or when other frictions constrain their ability to tailor incentive contracts, IIQ is unlikely to be the binding constraint. In such settings, these other frictions are more likely to already prevent boards from implementing more customized performance metrics, regardless of the quality of the internal information environment—weakening the relation between IIQ and performance metric selection. By contrast, when complexity, processing capacity, or external pressure increases the demand for customization and/or increases the scope for performance metric selection, IIQ becomes more likely to serve as the key limiting friction. In these settings, IIQ is more consequential, which we predict strengthens the relation between IIQ and performance metric selection.

To illustrate, consider situations in which IIQ is less consequential. For instance, if firms operate in an environment where relatively few performance metrics are optimal even in the absence of information constraints, IIQ is less likely to bind, as the unconstrained solution already falls within the feasible set. Similarly, if boards lack sufficient capacity to process and act on performance-related information, then that capacity itself becomes the limiting factor, rendering any improvement in IIQ less impactful. Finally, when firms face limited external environmental or social pressures—such as reputational scrutiny or stakeholder activism—the demand for more customized and externally aligned performance metrics is low, leaving little scope for IIQ to influence performance metric selection. By contrast, when firms are more complex, boards have greater capacity to process information, or external pressures are high, the role IIQ can play in influencing performance metric selection increases.

Collectively, the above discussion leads to our second hypothesis (stated in alternative form), which predicts that the association between higher quality internal information environments and the breadth and variety of performance metrics in a CEO's incentive contract varies in the cross-section of firms:

Hypothesis 2. The increase in the breadth and variety of performance metrics in a CEO's incentive contract in higher quality internal information environments is moderated by:

(a) organizational complexity;
(b) the ease of internal information processing; and
(c) external pressures.

3. Data and summary statistics

In this section, we discuss sample selection, key variable measurement, and present summary statistics. Appendix A details all variables. Appendix OA in the Online Appendix contains two proxy statement excerpts from Abbott Laboratories (2024) and Duke Energy Corporation (2024) that illustrate their CEO's incentive plan and how we calculate our main measures from that source data.

3.1. Data sources and sample construction

Data to create our sample are drawn from twelve sources: (1) ISS Incentive Lab for CEO incentive contract details; (2) ExecuComp for CEO equity holdings; (3) I/B/E/S, (4) Audit Analytics, and (5) LSEG ESG (formerly known as Refinitiv and ASSET4) for the dimensions of firms' IIQ; (6) Compustat for firm fundamentals; (7) Compustat Customer Segments for how many customer names the firm discloses; (8) CRSP and (9) Fama-French Portfolios and Factors for stock returns; (10) BoardEx for governance details; (11) RepRisk for firms' RepRisk Index scores for firms' reputation risk; and the (12) Hoberg-Phillips Data Library for the degree of product market competition (Hoberg and Phillips, 2010, 2016).

The sample begins in fiscal year 2006, following the SEC's (2006) mandate for firms to disclose information about their incentive-compensation contracts, and ends in fiscal year 2023, whereby we require future data on boards' choices for their executives' pay packages. The resulting sample consists of 20,409 firm-year observations.

3.2. Contract design details

Our main tests examine two aspects of the breadth and variety of performance metrics in CEOs' incentive contracts: (1) the number of performance metrics and (2) the dissimilarity of performance categories relative to industry peers (see Appendix OA in the Online Appendix for two example calculations). These two aspects complement each other in that each aspect provides different information about the design of CEOs' pay packages. In supplemental analyses, we also disaggregate both aspects into their financial and non-financial components.

With regard to the number of performance metrics, we take into account that incentive contracts can differ from each other in terms of their breadth. For example, two incentive contracts are not alike if one firm chooses to use only one performance metric and another firm chooses to use two performance metrics. To measure this dimension, we count the number of performance metrics that are given weight in the CEO's incentive contract. This count includes metrics from both the annual incentive plan and the long-term incentive plan, as reported in the ISS Incentive Lab database. To provide additional granularity, we also construct two related variables. *Number of Financial Metrics* counts the metrics in the following categories (i.e., the "*metricType*" field in the ISS Incentive Lab database): (1) accounting; (2) activity-related; (3) balance sheet-related; (4) cash flow; (5) earnings-related; (6) economic value; (7) investment ratios; (8) market-related; (9) revenue-related; (10) solvency-related; and (11) stock price. *Number of Non-Financial Metrics* counts the metrics in the following categories: (1) CSR; (2) environmental; (3) non-financial; (4) social; and (5) other.[12]

With regard to the dissimilarity of performance categories relative to industry peers, we further take into account that one firm's incentive contract may differ from rival firms' incentive contracts. This dimension provides further information about contract design if firms choose performance metrics belonging to different performance categories. For example, one firm might choose one financial performance metric and one social performance metric whereas another firm might choose one financial performance metric and one environmental performance metric (which is in a different category than a social one). Although both firms have two performance metrics and two performance categories, these incentive contracts are not alike. Therefore, we create a measure of contract dissimilarity that considers these differences. We label this variable *Metrics Dissimilarity*. To compute this variable, we first calculate for each industry-year the average number of performance metrics in each of the sixteen performance categories (i.e., the "*metricType*" field). The industry classification follows the 48 industry groups identified by Fama and French (1997). We then compute contract dissimilarity as the distance between the number of performance metrics in each performance category in the focal CEO's contract vis-à-vis the industry-year average contract:

$$\sum_{i=1}^{N=16} |f_i - a_i|, \tag{1}$$

where f_i is the focal CEO's number of performance metrics in performance category i, a_i is the industry-year average number of performance metrics in performance category i, and N is the total number of performance categories.[13][14]

In parallel, we also construct *Financial Metrics Dissimilarity* and *Non-Financial Metrics Dissimilarity*, which compute the same distance-based measure but restrict attention to the eleven financial categories and the five non-financial categories, respectively. These dissimilarity metrics allow us to assess how uniquely a firm structures its incentive contract within the financial and non-financial domains relative to its industry peers.

3.3. Internal information quality

A key construct in all our empirical analyses is the quality of firms' overall internal information environments. Previous work has used a variety of publicly available proxies to measure this construct, including the accuracy of management forecasts, the speed at which management releases an earnings announcement after fiscal year closing, audit fees, the absence of Sarbanes-Oxley Section 404 material weaknesses, and the absence of restatements.[15] These proxies are understood to reflect underlying aspects of firms' information acquisition processes, internal control effectiveness, and the design and functioning of internal information systems. However, it is an open question whether these measures also capture *non-financial* aspects of firms' internal environments. Although prior work suggests that financial and non-financial information aspects are complementary in supporting firms' strategic and operational goals (e.g., Simons, 1990; Kaplan and Norton, 1992; Chenhall, 2003), we broaden our measurement to better reflect this complementarity by supplementing the five financial proxies with two non-financial indicators drawn from LSEG ESG: environmental initiatives and ESG reporting scope.

A key advantage of relying on publicly available data is that it allows for broad coverage across firms and years, mitigating concerns

[12] The "Environmental" category includes clearly defined environmental performance metrics, such as "Greenhouse Gas Emissions," "Greenhouse Gas Intensity Reduction," and "Carbon Dioxide Emissions Reduction." By contrast, the "CSR" category includes the more vaguely defined ESG-based metrics that cannot be uniquely assigned to "Environmental" or "Social." For example, the category includes metrics labelled as "ESG," "Sustainability," and "Health Safety and Environmental."

[13] This distance measure, also known as Manhattan distance, calculates dissimilarity by summing the absolute differences across categories. Our results remain robust when we instead use Euclidean distance, which sums the squared differences, to measure dissimilarity across performance categories (untabulated).

[14] ISS Incentive Lab updated its categories for performance metrics on a few occasions over recent years to capture shifts in executive compensation practices. These changes include, for instance, separating financial metrics into accounting-based, price-based, and ratio-based categories, as well as adding distinct categories for environmental metrics. For example, beginning in 2018, ISS began to phase out the "Accounting" and "Stock Price" categories, replacing "Stock Price" with "Market-Related" and by more consistently filling the more granular categories such as "Balance Sheet-Related," "Cash Flow," "Earnings-Related, "Revenue-Related" rather than using the more aggregate "Accounting" category. Although these expansions increase the raw values for the *Metrics Dissimilarity* measure, they likely do not impact our results for two reasons. First, year fixed effects capture much of the temporal variation introduced by these updates. Second, the updates only reclassify metrics without affecting differences between firms that were using similar metrics. For instance, if two firms used comparable categories (e.g., an earnings and a stock price metric), their choices remain comparable after these database changes—as values for both firms will move in the same direction.

[15] See, for instance, Hogan and Wilkins (2008), Goodman et al. (2014), Gallemore and Labro (2015), and Heitzman and Huang (2019).

around data availability, sample selection, and generalizability that often arise with proprietary measures of IIQ (e.g., those based on executive surveys). To ensure that our inferences are not unique to one specific measure of information quality used by prior literature—but rather apply to this theoretical construct more generally—we construct an IIQ index that combines all proxies. Specifically, we define *IIQ* as the first principal component of the following seven measures of firms' IIQ: (1) *Forecast Accuracy*, which is the absolute difference between management's quarterly estimate of earnings per share and the firm's realized quarterly earnings per share; (2) *EA Speed*, which is the number of days between the end of the fiscal year and the firm's earnings announcement, divided by 365; (3) *Audit Intensity*, which is the firm's audit fees scaled by firm size; (4) *No SOX404 Weakness*, which indicates the absence of Sarbanes-Oxley Section 404 material weaknesses in internal controls; (5) *No Restatements*, which indicates the absence of restatements; (6) *Environmental Initiatives*, which indicates the presence of proactive environmental investments or expenditures to reduce future risks or increase future opportunities; and (7) *ESG Reporting Scope*, which indicates the presence of comprehensive environmental and social reporting. If necessary, we multiplied the variables by negative one so that larger values correspond to higher IIQ for every proxy. For each financial dimension, we compute the average value over the past three years. We then combine these values with the two non-financial indicators to construct a rolling index. This variable explains 24.906% of the total variance across all seven measures, with its eigenvalue exceeding 1 at $0.249 \times 7 \approx 1.743$.

3.4. Summary statistics

Table 1 presents summary statistics for the variables used in our analysis. Table 2 presents correlations between the variables used in our main analysis, with Spearman and Pearson correlation coefficients above and below the diagonal, respectively. Firms include, on average, 5.670 performance metrics in their CEO pay packages, with 3.882 financial metrics and 1.788 non-financial metrics. The average contract dissimilarity score is 4.441. The correlation between the number of metrics and dissimilarity measures ranges from 0.597 to 0.826, indicating that contracts with more metrics also tend to differ more from peer contracts in terms of metric composition. The average contract dissimilarity score implies that the average firm's contract differs $4.441/16 \times 100 \approx 27.756\%$ from rival firms' contracts within each performance category.

Another way to put the average contract dissimilarity score into context is by comparing it to the average of 5.670 performance metrics used in a typical CEO contract. At face value, this suggests that firms differ from the industry benchmark by approximately 78% of the number of metrics in the average contract (i.e., $4.441/5.670 \approx 78.325\%$). However, this interpretation can be misleading and should be treated with caution. The dissimilarity score reflects the total number of category-level differences—that is, how performance metrics are distributed across sixteen possible performance categories—rather than how many metrics are used in total. For example, two firms may each use five performance metrics, but if one firm focuses on financial metrics and the other on non-financial ones, their dissimilarity score may still be high despite having the same total count. Conversely, a firm with seven metrics could have a low dissimilarity score if its category usage closely matches the industry average. Therefore, although the dissimilarity score is numerically comparable to the number of metrics, it captures a fundamentally different dimension of contract design: variation in metric composition, not just metric count.

Summary statistics for our composite measure of IIQ (*IIQ*) are relatively uninformative because it is a principal component score with a mean of zero and a standard deviation of one. To provide descriptive context for the main variables used in our analysis, Fig. 1 plots the evolution of the average values of firms' IIQ, alongside the number of performance metrics and the dissimilarity of the focal CEO's incentive contract relative to industry peers, over time. Although all three variables exhibit upward trends, these patterns are presented for descriptive purposes only and are not used for identification, as our empirical specifications include year (and firm) fixed effects. Fig. 2 disaggregates the evolution of performance metrics and contract dissimilarity into financial and non-financial components, showing that both dimensions contribute to overall trends, although non-financial components are rising faster in terms of growth rate. Finally, Fig. 3 plots the evolution of the average values of the seven proxies underlying firms' IIQ over time, where larger values represent better IIQ for all seven proxies.[16] Notably, the non-financial components show more pronounced increases in recent years. The patterns in Figs. 2 and 3, which are generally more pronounced for non-financial components, are consistent with the rising attention to the non-financial aspect of firms' performance (e.g., ESG) in recent years. Together, these figures provide descriptive context on how the key variables evolve over the sample period, including the growing use of non-financial performance metrics in recent years.

4. Main empirical tests and results

4.1. Internal information quality and performance metric selection

In this section, we test Hypothesis 1 by examining how firms' internal information environment relates to executives' contract design details by estimating equations of the form:

$$[Contract_{it+1}] = \beta \bullet IIQ_{it} + \Phi \bullet X_{it} + \mathrm{T} \bullet \tau_i + \Omega \bullet \xi_t + \varepsilon_{it+1}, \tag{2}$$

[16] Recently, Incentive Lab expanded its coverage from mostly S&P 500 firms to S&P 1500 firms with the extended coverage being maintained from 2014 onwards, so some proxies show sharper changes in 2014.

Table 1
Summary statistics.

	Mean	Std. Dev.	10th	25th	50th	75th	90th
Number of Metrics	5.670	4.433	1.000	3.000	5.000	7.000	10.000
Number of Financial Metrics	3.882	2.707	1.000	2.000	4.000	5.000	7.000
Number of Non-Financial Metrics	1.788	2.961	0.000	0.000	1.000	2.000	5.000
Metrics Dissimilarity	4.441	3.734	1.152	2.286	3.806	5.714	8.220
Financial Metrics Dissimilarity	2.644	2.203	0.444	1.263	2.286	3.474	5.000
Non-Financial Metrics Dissimilarity	1.350	2.422	0.000	0.000	0.625	1.860	3.587
IIQ	0.004	0.974	−1.084	−0.511	0.005	0.512	1.209
Forecast Accuracy	−0.353	0.683	−1.080	−0.428	0.000	0.000	0.000
EA Speed	−0.112	0.036	−0.158	−0.142	−0.109	−0.081	−0.067
Audit Simplicity	−0.191	0.203	−0.395	−0.231	−0.129	−0.073	−0.040
No SOX404 Weakness	0.960	0.147	1.000	1.000	1.000	1.000	1.000
No Restatement	0.907	0.202	0.667	1.000	1.000	1.000	1.000
Environmental Initiatives	0.091	0.287	0.000	0.000	0.000	0.000	0.000
ESG Reporting Scope	0.298	0.457	0.000	0.000	0.000	1.000	1.000
Delta	5.482	1.627	3.539	4.462	5.485	6.498	7.427
Vega	2.904	2.430	0.000	0.000	3.438	5.015	5.941
Age	4.054	0.120	3.892	3.989	4.060	4.127	4.190
Tenure	2.340	0.686	1.386	1.946	2.398	2.773	3.219
Number of Rivals	3.391	1.565	1.386	2.197	3.296	4.522	5.835
Rival Similarity	0.027	0.029	0.004	0.011	0.018	0.030	0.064
Sales	7.800	1.507	5.903	6.788	7.800	8.881	9.847
Book-to-Market	0.633	0.279	0.266	0.417	0.626	0.846	0.986
Leverage	0.243	0.217	0.001	0.069	0.213	0.350	0.503
Sales Growth	0.156	2.083	−0.105	−0.009	0.065	0.158	0.313
Cash	0.152	0.173	0.012	0.033	0.090	0.204	0.383
CEO Duality	0.409	0.492	0.000	0.000	0.000	1.000	1.000
Board Size	2.345	0.226	2.079	2.197	2.398	2.485	2.639
Board Independence	0.714	0.139	0.538	0.615	0.700	0.857	0.889
Board Experience	0.065	0.098	0.000	0.000	0.000	0.111	0.200

This table presents summary statistics for the variables used in our main analysis. Appendix A defines all variables.

where the indices i and t correspond to firm and time, respectively.

[*Contract*] is one of the two measures of executives' contract design details (see Section 3.2 for details). *IIQ* is our composite measure of firms' overall IIQ (see Section 3.3 for details). X is a vector of control variables, which we discuss below. All independent variables are measured prior to the dependent variable to capture the most recent factors contemporaneous with boards' choices for executive incentive contracts. To correct for any arbitrary correlation in the firm-year specific error term ε_{it+1} across time within a given firm, we base inferences on standard errors clustered at the level of treatment (i.e., firm) following Abadie et al. (2023).

Hypothesis 1 provides the testable prediction that *on average* and *within a given firm*, time-series variation in IIQ is positively associated with time-series variation in performance metric selection. This prediction delineates that β in Eq. (2) should be positive. It also implies that this test should consist of an analysis that relies on a comparison within firms, implying the inclusion of firm fixed effects τ_i (and year fixed effects ξ_t to rule out common time trends unrelated to our prediction). This within-firm design has an added benefit that it helps to reduce concerns that omitted time-invariant firm-specific characteristics drive our results. To assess the robustness and interpretability of our findings, we also estimate variants of Eq. (2) using alternative fixed effects specifications. These include models with only firm fixed effects and models that replace all fixed effects with industry fixed effects (based on the 48 Fama and French (1997) industry classifications).

We also include in X an array of executive-year, firm-year, and governance-year time-varying controls for which there exists economic theory or empirical evidence to support a relation with our key constructs. Executive-level controls are the CEO's equity portfolio incentives (*Delta* and *Vega*), his/her age (*Age*), and his/her tenure (*Tenure*) (Core and Guay, 1999; Guay, 1999). Firm-level controls include the number of product market competitors and their mean similarity with the focal firm, based on the Text-based Network Industry Classifications (TNIC) data from Hoberg and Phillips (2010, 2016) (*Number of Rivals* and *Rival Similarity*), and also include the annual revenue (*Sales*), book-to-market ratio (*Book-to-Market*), leverage (*Leverage*), sales growth (*Sales Growth*), and cash and cash equivalents balance (*Cash*). Governance-level controls include whether the CEO is also the Chairman of the Board (*CEO Duality*), the number of board members (*Board Size*), the fraction of board members who are independent (*Board Independence*), and the fraction of board members who sit on at least three other boards (*Board Experience*) (Linck et al., 2008; Field et al., 2013).

Table 3 presents the results from estimating the relation between IIQ and details of CEOs' incentive contracts, with *Number of*

Table 2
Correlation matrix.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
(1)		0.60	0.25	−0.06	−0.07	0.05	−0.02	0.09	0.09	0.19	0.09	0.15	−0.05	−0.11	−0.03	0.21	0.13	0.12
(2)	0.83		0.20	−0.09	−0.24	0.10	0.02	0.08	0.08	0.09	0.10	0.14	0.00	−0.10	−0.06	0.10	0.17	0.07
(3)	0.23	0.18		0.22	0.11	0.08	0.21	−0.04	0.00	0.65	−0.01	0.16	−0.04	−0.13	0.10	0.41	−0.15	0.13
(4)	−0.03	−0.05	0.21		0.50	0.18	0.37	0.02	−0.01	0.34	−0.34	0.02	0.13	0.10	0.31	0.13	−0.29	0.06
(5)	−0.09	−0.19	0.07	0.45		−0.02	0.11	0.01	−0.02	0.27	−0.12	−0.01	−0.07	0.08	0.18	0.15	−0.23	0.06
(6)	0.05	0.08	0.08	0.18	−0.03		0.31	0.02	0.04	0.09	0.07	0.01	−0.03	−0.06	0.24	0.06	0.01	−0.02
(7)	0.00	0.02	0.20	0.39	0.10	0.32		0.00	0.02	0.17	0.00	−0.03	−0.02	−0.07	0.24	0.12	−0.08	−0.02
(8)	0.11	0.09	−0.06	0.01	0.01	0.02	0.00		0.67	−0.12	0.18	−0.14	0.07	0.06	−0.01	0.05	−0.01	0.02
(9)	0.09	0.07	−0.05	−0.05	−0.04	0.04	0.01	0.60		−0.06	0.22	−0.07	0.04	−0.03	0.01	0.09	−0.02	0.02
(10)	0.17	0.10	0.65	0.32	0.23	0.09	0.17	−0.14	−0.18		0.07	0.29	−0.07	−0.15	0.13	0.50	−0.32	0.21
(11)	0.08	0.07	0.00	−0.33	−0.11	0.07	−0.01	0.18	0.22	0.08		0.06	−0.22	−0.36	0.03	0.16	0.13	0.04
(12)	0.08	0.09	0.12	0.02	−0.01	0.01	−0.04	−0.09	−0.10	0.24	0.01		−0.07	−0.31	−0.01	0.13	0.01	0.17
(13)	0.00	0.00	−0.07	−0.01	−0.01	−0.01	−0.04	0.05	0.06	−0.07	−0.04	−0.01		0.04	−0.02	−0.09	−0.03	−0.02
(14)	−0.08	−0.06	−0.18	0.08	0.05	−0.06	−0.08	0.18	0.05	−0.24	−0.32	−0.08			−0.06	−0.18	−0.06	−0.05
(15)	0.00	−0.02	0.10	0.30	0.17	0.24	0.24	−0.01	0.02	0.13	0.02	−0.03	−0.01	−0.07		0.08	−0.06	0.03
(16)	0.17	0.09	0.37	0.12	0.14	0.05	0.10	0.07	0.12	0.47	0.14	0.09	−0.04	−0.17	0.07		−0.05	0.18
(17)	0.06	0.10	−0.13	−0.28	−0.22	0.01	−0.07	0.01	0.07	−0.30	0.12	−0.01	0.01	−0.04	−0.06	−0.04		0.00
(18)	0.09	0.06	0.08	0.03	0.04	−0.02	−0.01	0.04	0.01	0.16	0.04	0.13	0.01	−0.01	0.02	0.11	0.02	

Legend: (1) *Number of Metrics*; (2) *Metrics Dissimilarity*; (3) *IIQ*; (4) *Delta*; (5) *Vega*; (6) *Age*; (7) *Tenure*; (8) *Number of Rivals*; (9) *Rival Similarity*; (10) *Sales*; (11) *Book-to-Market*; (12) *Leverage*; (13) *Sales Growth*; (14) *Cash*; (15) *CEO Duality*; (16) *Board Size*; (17) *Board Independence*; and (18) *Board Experience*.
This table presents a correlation matrix for the variables used in our main analysis. Spearman correlation coefficients appear above the diagonal, and Pearson correlation coefficients appear below it. Appendix A defines all variables.

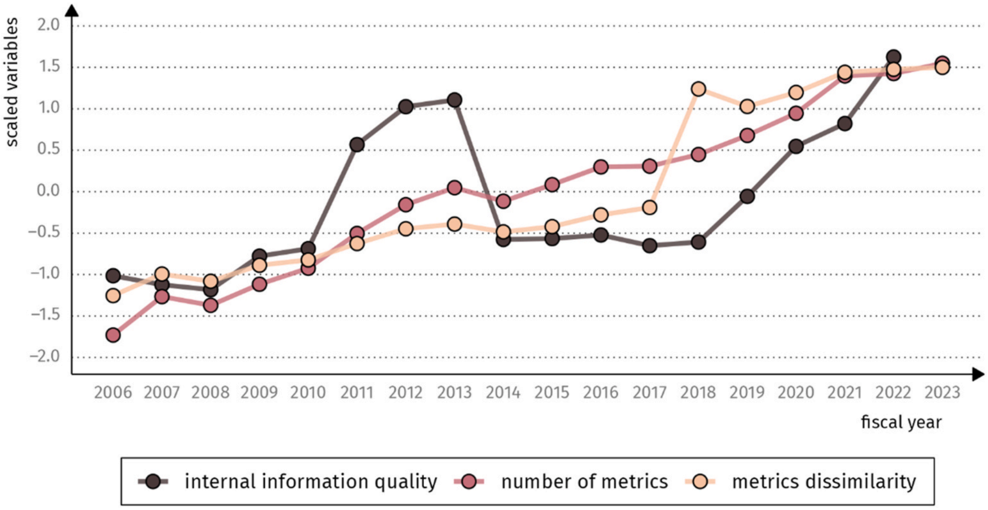

Fig. 1. Internal information quality and performance metric selection over time.
This figure illustrates the evolution of the average values of firms' internal information quality, the number of performance metrics and the dissimilarity of the focal CEO's incentive contract relative to industry peers over time. We standardize all three variables using z-transformations to ensure they are comparable in the plot. Appendix A defines all variables.

Metrics and *Metrics Dissimilarity* as the dependent variable in columns (1) to (3) and columns (4) to (6), respectively. Across all specifications, we find that the coefficient on *IIQ* is positive.[17] This pattern suggests that both across firms and within firms, variation in IIQ is positively associated with the number of performance metrics and the dissimilarity in performance categories relative to industry peers.[18]

Although we recognize that both IIQ and performance metric selection are endogenous firm choices, the estimated coefficients suggest a meaningful economic association between the two. Specifically, within a given firm, a one standard deviation increase in IIQ is associated with an increase in the number of performance metrics equal to 12.205% of the within-firm standard deviation of that variable, and an increase in contract dissimilarity equal to 12.033% of its within-firm standard deviation.[19] Based on between-firm variation, the corresponding effect sizes are 14.689% and 13.781%, respectively. Thus, as internal information quality increases, so do the number and dissimilarity of performance metrics. This relation could reflect that better IIQ makes it more likely for boards to increase the breadth and variety of performance metrics in their design of CEO incentive contracts (as predicted by Hypothesis 1), or that firms enhance their internal performance measurement systems in ways that both increase the number and dissimilarity of metrics as well as improve IIQ.

We acknowledge that if such a correlated omitted variable simultaneously affects firms' IIQ and the propensity for boards to alter their executives' pay packages, then these estimates may be biased and need not reflect the effect of IIQ on performance metric selection.[20] To sharpen our inferences, we therefore turn to a quasi-experimental design that examines how performance metric selection

[17] To caveat, the positive association we document should not be interpreted as evidence that ever-higher IIQ will lead firms to adopt performance metrics across all metric categories. Rather, our findings suggest that internal information systems act as a friction, which mainly determines the supply side of the demand-supply equilibrium for the final performance metric selection. As this friction diminishes, firms can move closer to their unconstrained optimal contract—one that aligns with their strategic objectives and monitoring needs, which likely does not include metrics across all categories. In other words, the demand for performance metrics will act as a limiting factor for an increase in the supply side of the equilibrium.

[18] The sharp reduction in the coefficient on *IIQ* once year fixed effects are included suggests a strong temporal component in the association we examine. This reflects both the causal effect of overall improvements in IIQ—especially for non-financial data, as shown in Fig. 3—and broader time trends, such as the rising emphasis on non-financial performance, which simultaneously influence information quality and executive incentive contracts. Although we highlight this temporal pattern as it reflects a major shift in the business environment, our core inferences rely on specifications with year fixed effects to mitigate endogeneity concerns.

[19] To interpret the magnitude of these association, we follow Mummolo and Peterson (2018), Mitton (2024), and Breuer and deHaan (2024), and express effect sizes relative to the within-fixed-effects standard deviation of the dependent variable. Specifically, we multiply the estimated coefficient on *IIQ* by the standard deviation of *IIQ* after removing fixed effects, and scale this product by the standard deviation of the dependent variable (also net of fixed effects).

[20] The accounting treatment for stock-based compensation changed in 2005, right before our sample period. Prior research shows that accounting rules can influence the structure of executive compensation contracts and the choice of performance metrics used in incentive plans (e.g., Carter et al., 2007; Cadman et al., 2013). We discuss the accounting treatment and any potential impact of that change through long-term incentive policies that may have lingered into our sample period in the Online Appendix. To alleviate concerns from that potential impact, we also check in Table OA1 that our main findings in Table 3 are robust to restricting the sample period to start after 2009, because such long-term incentive policies would have been periodically reviewed and updated.

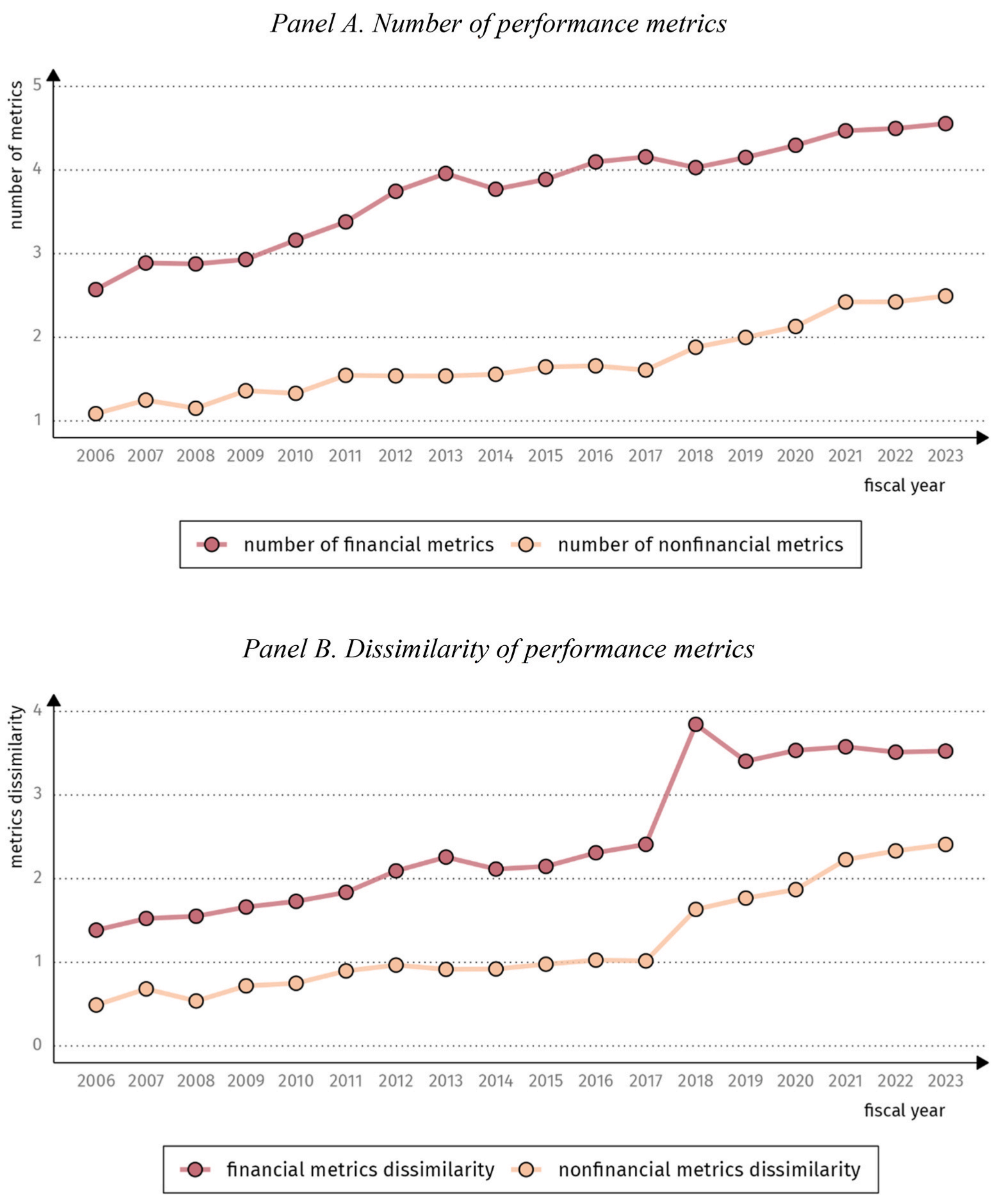

Fig. 2. Performance metric selection over time—by type.
This figure illustrates the evolution of the average values of firms' number of performance metrics and the dissimilarity of the focal CEO's incentive contract relative to industry peers over time, separately for financial and non-financial performance metrics. Appendix A defines all variables.

responds to two exogenous shocks to firms' internal information environment (see Section 5 for details). Before doing so, however, we first explore settings in which more tailored or complex performance metrics are likely to be demanded and settings that increase the scope for performance metric selection, as predicted by Hypothesis 2. This allows us to test whether IIQ plays a differential facilitating or constraining role in enabling more tailored performance metric structures when such complexity is needed.

4.2. *Internal information quality and performance metric selection—cross-sectional variation*

In the next subsections, we test Hypothesis 2 by examining whether the relation between IIQ and performance metric selection varies across firm settings with differing demands for tailored performance metrics. Specifically, we analyze three types of settings in which such demand is likely to arise: (1) firms characterized by greater organizational complexity; (2) firms for which internal information is easier to process and act upon; and (3) firms facing heightened external environmental and social pressures. These factors are conceptually distinct from those that directly influence the quality of firms' internal information environment. Rather, they

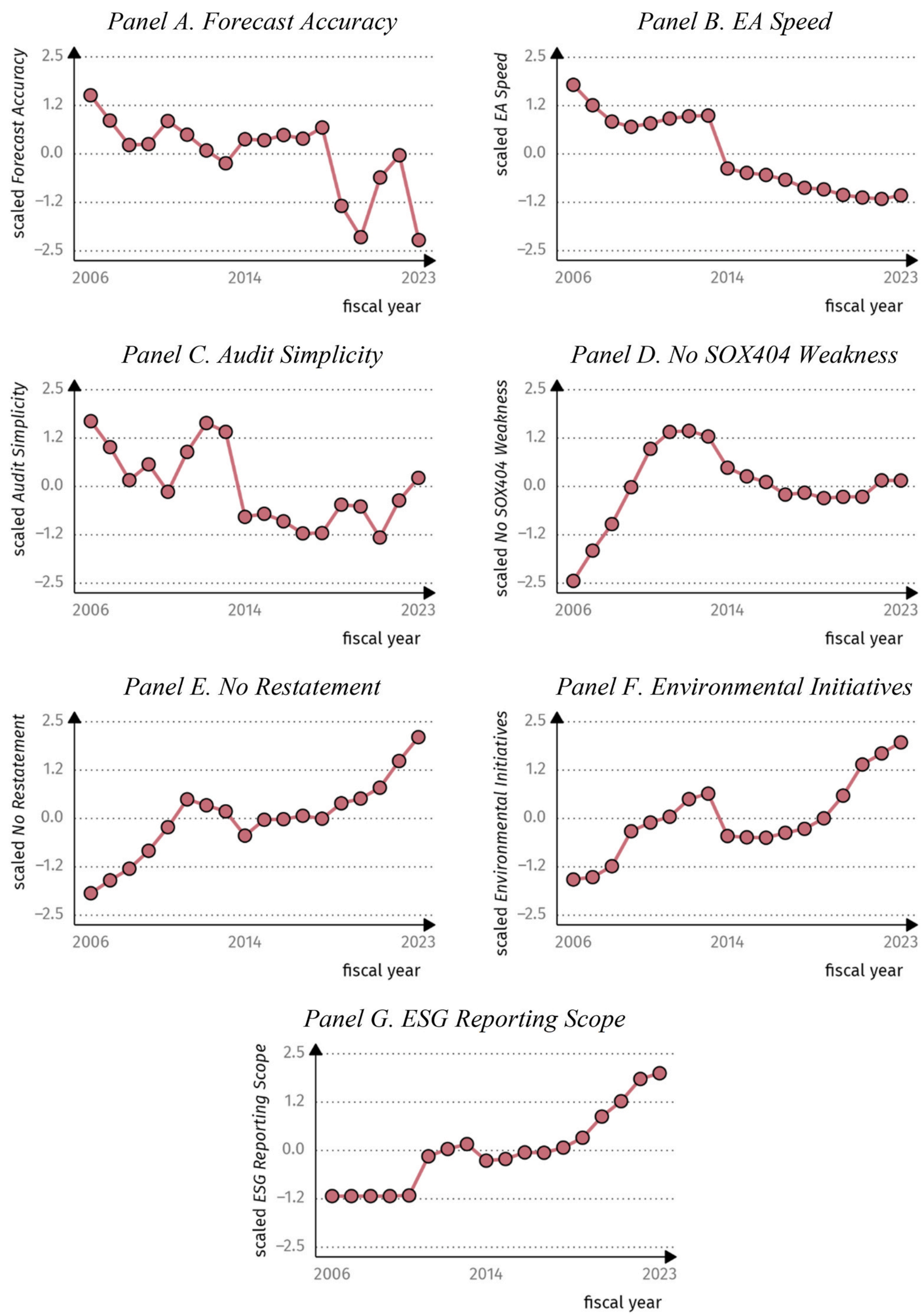

Fig. 3. Internal information quality over time—by type.
This figure illustrates the evolution of the average values of the seven proxies underlying firms' internal information quality over time. We standardize all seven variables using *z*-transformations to ensure they are comparable in the plots. Appendix A defines all variables.

determine the demand or need for more customized metrics: for instance, by increasing the complexity of firms' operations, the interpretability of relevant information, or the external pressures to address certain stakeholder concerns. To test these predictions, we construct variables for each cross-sectional factor and estimate Eq. (2) separately for the subsamples with relatively low and high values of each variable.

Table 3
Internal information quality and performance metric selection.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable:					
	Number of Metrics$_{t+1}$			*Metrics Dissimilarity*$_{t+1}$		
IIQ	0.701***	0.701***	0.192***	0.558***	0.678***	0.158***
	(0.077)	(0.077)	(0.074)	(0.059)	(0.065)	(0.060)
Delta	−0.059	−0.036	−0.028	0.041	−0.013	−0.010
	(0.059)	(0.045)	(0.043)	(0.048)	(0.042)	(0.037)
Vega	−0.165***	−0.199***	−0.029	−0.283***	−0.263***	−0.034
	(0.029)	(0.025)	(0.028)	(0.023)	(0.023)	(0.025)
Age	1.225**	1.693***	0.325	1.619***	2.255***	0.541
	(0.499)	(0.545)	(0.499)	(0.387)	(0.485)	(0.404)
Tenure	−0.255***	−0.027	−0.029	−0.083	0.014	0.018
	(0.083)	(0.105)	(0.098)	(0.065)	(0.096)	(0.085)
Number of Rivals	0.083	0.017	0.162**	−0.050	−0.113	0.086
	(0.054)	(0.076)	(0.070)	(0.039)	(0.070)	(0.061)
Rival Similarity	7.227**	4.864**	3.973*	0.583	2.417	1.829
	(2.884)	(1.968)	(2.049)	(2.073)	(1.653)	(1.564)
Sales	0.428***	0.583***	−0.054	0.243***	0.738***	−0.109
	(0.070)	(0.104)	(0.110)	(0.053)	(0.088)	(0.085)
Book-to-Market	−0.098	0.095	0.398	−0.030	0.129	0.447**
	(0.259)	(0.252)	(0.242)	(0.192)	(0.248)	(0.227)
Leverage	0.536	1.352***	−0.646**	0.568**	1.909***	−0.606**
	(0.358)	(0.329)	(0.324)	(0.244)	(0.293)	(0.269)
Sales Growth	0.005	−0.031*	−0.019	0.005	−0.026***	−0.016**
	(0.010)	(0.016)	(0.014)	(0.008)	(0.009)	(0.008)
Cash	−0.800	0.132	−0.291	−0.130	−0.228	−0.461
	(0.582)	(0.615)	(0.597)	(0.413)	(0.572)	(0.548)
CEO Duality	−0.065	−0.184*	0.275***	−0.224**	−0.309***	0.244***
	(0.133)	(0.100)	(0.100)	(0.102)	(0.093)	(0.086)
Board Size	0.891***	−0.081	0.338	0.092	−0.046	0.246
	(0.296)	(0.372)	(0.357)	(0.217)	(0.352)	(0.329)
Board Independence	2.655***	2.702***	−0.903**	2.524***	2.211***	−0.912**
	(0.408)	(0.386)	(0.454)	(0.303)	(0.340)	(0.376)
Board Experience	0.334	−0.356	−0.094	−0.364	−0.243	−0.045
	(0.662)	(0.615)	(0.573)	(0.493)	(0.552)	(0.471)
Fixed effects	industry	firm	firm, year	industry	firm	firm, year
Observations	20,409	20,409	20,409	20,409	20,409	20,409
Adjusted R^2	15.308%	47.475%	50.401%	15.202%	37.881%	45.559%

This table presents results from estimating the relation between internal information quality and details of CEOs' incentive contracts. Columns (1) through (3) present results for the number of performance metrics (*Number of Metrics*), with the three columns presenting, respectively, results from specifications with industry, firm, and firm and year fixed effects. Columns (4) through (6) present analogous results for the dissimilarity of the focal CEO's incentive contract relative to industry peers (*Metrics Dissimilarity*). For brevity, we do not report their coefficient estimates and standard errors of the various fixed effects. Standard errors are in parentheses and are adjusted for within-cluster correlation at the level of treatment (i.e., firm). *, **, and *** denote that the coefficient is significantly different from zero at, respectively, two-tailed probability levels of 10%, 5%, and 1%. Appendix A defines all variables.

4.2.1. Cross-sectional variation in organizational complexity

We examine whether the relation between firms' IIQ and details of CEOs' incentive contracts varies with organizational complexity. We measure organizational complexity using *Segments* and *σReturns*, with larger values of each measure corresponding to greater organizational complexity. *Segments* is the number of business segments the firm operates in (with data coming from Compustat Segments). *σReturns* is the firm's historical idiosyncratic stock return volatility, defined as the standard deviation of the firm's annualized returns that are unexplained by the Carhart (1997) four-factor model estimated over the prior 60 months (with data coming from CRSP and Fama-French Portfolios and Factors).

Table 4 presents the results, with Panel A presenting the results after partitioning on *Segments*, and Panel B presenting the results after partitioning on *σReturns*. Across both panels, we find that the coefficient on *IIQ* is economically and statistically larger for firms with greater organizational complexity—those with more business segments or higher idiosyncratic return volatility—relative to their less complex counterparts. These results are consistent with our prediction that IIQ plays a more prominent role in enabling tailored performance metric design in more complex organizational settings. When complexity increases the demand for more tailored contracts, limited IIQ becomes a more binding constraint for boards' performance metric selection.

Table 4
Internal information quality and performance metric selection—cross-sectional variation in organizational complexity.

Panel A. Number of business segments

Subsample—split by *Segments*	(1)	(2)	(3)	(4)
	low	high	low	high
Variable	Dependent variable:			
	Number of Metrics$_{t+1}$		*Metrics Dissimilarity*$_{t+1}$	
IIQ	0.089	0.464***	0.092	0.338**
	(0.083)	(0.162)	(0.067)	(0.132)
Controls	yes	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year	firm, year
Observations	15,202	5,179	15,202	5,179
Adjusted R^2	52.469%	49.181%	48.638%	41.176%
Difference in *IIQ*	0.375**		0.246**	

Panel B. Idiosyncratic stock return-volatility

Subsample—split by *σReturns*	(1)	(2)	(3)	(4)
	low	high	low	high
Variable	Dependent variable:			
	Number of Metrics$_{t+1}$		*Metrics Dissimilarity*$_{t+1}$	
IIQ	0.074	0.283***	0.055	0.282***
	(0.104)	(0.108)	(0.089)	(0.087)
Controls	yes	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year	firm, year
Observations	9,809	9,808	9,809	9,808
Adjusted R^2	51.068%	51.810%	46.755%	44.688%
Difference in *IIQ*	0.208*		0.227**	

This table presents results from examining whether the within-firm relation between internal information quality and details of CEOs' incentive contracts varies with organizational complexity. We estimate the regression models separately for subsamples split by the median value of the cross-sectional variables, allowing all coefficients and fixed effects to vary across the two subsamples. We test for signed differences in the coefficient estimates between the two subsamples using *Z*-tests and report the results at the bottom of each panel. The regression models also include firm and year effects and *Delta*, *Vega*, *Age*, *Tenure*, *Number of Rivals*, *Rival Similarity*, *Sales*, *Book-to-Market*, *Leverage*, *Sales Growth*, *Cash*, *CEO Duality*, *Board Size*, *Board Independence*, and *Board Experience*; for brevity, we do not report their coefficient estimates and standard errors. Standard errors are in parentheses and are adjusted for within-cluster correlation at the level of treatment (i.e., firm). *, **, and *** denote that the coefficient is significantly different from zero at, respectively, two-tailed probability levels of 10%, 5%, and 1%. Appendix A defines all variables.

4.2.2. Cross-sectional variation in ease of internal information processing

Next, we examine whether the relation between firms' IIQ and details of CEOs' incentive contracts varies with ease of internal information processing. We measure ease of internal information processing using *Board Experience* and *Comovement*, with larger values of each measure corresponding to greater ease of internal information processing. *Board Experience* is the fraction of board members who sit on at least three other boards (with data coming from BoardEx). *Comovement* is the correlation between quarterly stock returns and earnings per share estimated over a 20-quarter period (with data coming from Compustat Quarterly), which makes processing of multiple metrics easier for the board as there is less idiosyncratic fluctuation in the metrics that needs to be processed.[21]

Table 5 presents the results, with Panel A presenting the results after partitioning on *Board Experience*, and Panel B presenting the results after partitioning on *Comovement*. Across both panels, we find that the coefficient on *IIQ* is in general economically and statistically larger for firms with greater ease of internal information processing—those with more experienced boards and greater performance comovement—relative to their counterparts with less processing capacity. These findings are consistent with our prediction that IIQ is more likely to influence performance metric design when boards are better equipped to process and use that information. This suggests that even if high quality information is available from firms' internal information system, its value for contract design depends on boards' capacity to absorb and apply it.

[21] In a similar vein, Albuquerque et al. (2025) find that the negative performance effects of compensation contract complexity are mitigated when performance metrics are more highly correlated, resulting in reduced information processing costs.

Table 5
Internal information quality and performance metric selection—cross-sectional variation in ease of internal information processing.

Panel A. Experienced boards

Subsample—split by *Board Experience*	(1)	(2)	(3)	(4)
	low	high	low	high
Variable	Dependent variable:			
	Number of Metrics$_{t+1}$		*Metrics Dissimilarity*$_{t+1}$	
IIQ	0.086	0.294**	0.121	0.192*
	(0.091)	(0.124)	(0.075)	(0.100)
Controls	yes	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year	firm, year
Observations	11,863	8,546	11,863	8,546
Adjusted R^2	48.459%	52.816%	43.430%	48.843%
Difference in *IIQ*	0.208*		0.071	

Panel B. Comovement between financial performance metrics

Subsample—split by *Comovement*	(1)	(2)	(3)	(4)
	low	high	low	high
Variable	Dependent variable:			
	Number of Metrics$_{t+1}$		*Metrics Dissimilarity*$_{t+1}$	
IIQ	0.001	0.369***	0.024	0.315***
	(0.109)	(0.096)	(0.091)	(0.077)
Controls	yes	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year	firm, year
Observations	9,808	9,807	9,808	9,807
Adjusted R^2	49.849%	54.123%	43.423%	50.886%
Difference in *IIQ*	0.369***		0.291***	

This table presents results from examining whether the within-firm relation between internal information quality and details of CEOs' incentive contracts varies with ease of internal information processing. We estimate the regression models separately for subsamples split by the median value of the cross-sectional variables, allowing all coefficients and fixed effects to vary across the two subsamples. We test for signed differences in the coefficient estimates between the two subsamples using *Z*-tests and report the results at the bottom of each panel. The regression models also include firm and year effects and *Delta*, *Vega*, *Age*, *Tenure*, *Number of Rivals*, *Rival Similarity*, *Sales*, *Book-to-Market*, *Leverage*, *Sales Growth*, *Cash*, *CEO Duality*, *Board Size*, *Board Independence*, and *Board Experience*; for brevity, we do not report their coefficient estimates and standard errors. Standard errors are in parentheses and are adjusted for within-cluster correlation at the level of treatment (i.e., firm). *, **, and *** denote that the coefficient is significantly different from zero at, respectively, two-tailed probability levels of 10%, 5%, and 1%. Appendix A defines all variables.

4.2.3. *Cross-sectional variation in external environmental and social pressure*

We examine whether the relation between firms' IIQ and details of CEOs' incentive contracts varies with external environmental and social pressure. We measure external environmental and social pressure using *E-Risk* and *S-Risk*, with larger values of each measure corresponding to greater external pressure. *E-Risk* and *S-Risk* are, respectively, the firm's environmental and social reputation risk (with data coming from RepRisk).[22]

Table 6 presents the results, with Panel A presenting the results after partitioning on *E-Risk*, and Panel B presenting the results after partitioning on *S-Risk*. In Panel A, we find that the coefficient on *IIQ* is statistically and economically larger for firms facing higher environmental reputation risk when examining contract dissimilarity. We find no statistically significant difference in the coefficient on *IIQ* when examining the number of performance metrics, although the coefficients for the low pressure subsample are generally smaller in magnitude. In contrast, Panel B shows that the coefficient on *IIQ* is statistically and economically larger for firms facing higher social reputation risk. These results suggest that IIQ plays a more prominent role in influencing performance metric selection in settings where firms are under greater environmental or social pressure. The finding that stronger associations arise primarily when external pressures heighten the demand for tailored incentives suggests that IIQ constrains firms' capacity to adjust performance metrics in response.

5. Quasi-experimental tests and results

Our results to this point lend themselves to at least three possible interpretations. First, high quality internal information

[22] The RepRisk Index is a proprietary algorithm developed by RepRisk that dynamically captures and quantifies a company's or project's reputational risk exposure to ESG issues (RepRisk, 2023). The index is performance-based and has been used in previous work to proxy for firms' ESG performance. See, for instance, Li and Wu (2020) and Gantchev et al. (2022).

Table 6
Internal information quality and performance metric selection—cross-sectional variation in external environmental and social pressure.

Panel A. Environmental reputation risk				
Subsample—split by *E-Risk*	(1)	(2)	(3)	(4)
	low	high	low	high
Variable	Dependent variable:			
	Number of Metrics$_{t+1}$		*Metrics Dissimilarity*$_{t+1}$	
IIQ	0.094	0.331*	0.124	0.353**
	(0.110)	(0.173)	(0.091)	(0.142)
Controls	yes	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year	firm, year
Observations	9,140	4,257	9,140	4,257
Adjusted R^2	44.405%	48.536%	39.965%	42.302%
Difference in *IIQ*	0.238		0.228*	

Panel B. Social reputation risk				
Subsample—split by *S-Risk*	(1)	(2)	(3)	(4)
	low	high	low	high
Variable	Dependent variable:			
	Number of Metrics$_{t+1}$		*Metrics Dissimilarity*$_{t+1}$	
IIQ	0.027	0.291**	0.040	0.313***
	(0.143)	(0.140)	(0.122)	(0.113)
Controls	yes	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year	firm, year
Observations	7,182	6,215	7,182	6,215
Adjusted R^2	44.591%	49.577%	38.920%	43.140%
Difference in *IIQ*	0.264*		0.273*	

This table presents results from examining whether the within-firm relation between internal information quality and details of CEOs' incentive contracts varies with external environmental and social pressure. We estimate the regression models separately for subsamples split by the median value of the cross-sectional variables, allowing all coefficients and fixed effects to vary across the two subsamples. We test for signed differences in the coefficient estimates between the two subsamples using *Z*-tests and report the results at the bottom of each panel. The regression models also include firm and year effects and *Delta*, *Vega*, *Age*, *Tenure*, *Number of Rivals*, *Rival Similarity*, *Sales*, *Book-to-Market*, *Leverage*, *Sales Growth*, *Cash*, *CEO Duality*, *Board Size*, *Board Independence*, and *Board Experience*; for brevity, we do not report their coefficient estimates and standard errors. The sample size in these cross-sectional analyses differs from the preceding analyses because reputation risk data is only available until 2020. Standard errors are in parentheses and are adjusted for within-cluster correlation at the level of treatment (i.e., firm). *, **, and *** denote that the coefficient is significantly different from zero at, respectively, two-tailed probability levels of 10%, 5%, and 1%. Appendix A defines all variables.

environments help boards to design more tailored and effective executive compensation packages.[23] Second, the causal direction could run in reverse: changes to executive incentive contracts might prompt improvements in firms' internal information systems. Third, both IIQ and performance metric selection may be jointly determined by an unobserved factor, reflecting the endogenous nature of both choices. We believe that distinguishing between these explanations provides greater insights into the relation between IIQ and performance metric selection.

For that aim, we examine the relation between IIQ and performance metric selection using two shocks to firms' internal information environment. The goal of these analyses is not to achieve full causal identification, but rather to leverage plausibly exogenous variation that enhances the credibility of our empirical design. Although we acknowledge that firms invest in internal reporting systems for a range of reasons beyond regulatory changes, these shocks offer a useful setting to explore the facilitating role of internal information systems in shaping performance metric selection. Specifically, the accounting standards we study are major and require firms to upgrade their internal reporting infrastructure and collect data that may have previously been considered too costly or unnecessary. As a result, the marginal cost of defining, collecting, validating, and implementing performance metrics declines, even if firms had no explicit intention to adjust their incentive contracts.

The two regulatory events we analyze are the announcements of Accounting Standards Update (ASU) No. 2014-09—Revenue from

[23] This causal interpretation does not need to assume an exogenous change in IIQ as the starting point of the entire causal chain. In most cases, firms' strategic policy decisions or other reasons for a heightened demand for performance metrics in the incentive contract will trigger changes in internal information systems, which in turn affect the design of incentive contracts. The distinction of this causal interpretation from the correlated omitted variable concern is that firms' strategic policy decisions causally affect performance metric selection *through* IIQ in the former but not in the latter, while being a crucial factor in both scenarios. Thus, in the former scenario, IIQ can be viewed as the necessary condition for the changes in the performance metrics, which makes it a causal determinant of incentive contracts at least narrowly in the final part of the causal chain. We fully acknowledge that our findings can be driven by more fundamental forces that we leave for future research.

Contracts with Customers (codified as ASC 606)—and ASU No. 2016-02—Leases (codified as ASC 842). ASC 606 is about contracts with customers to transfer goods or services (FASB, 2014).[24] ASC 842 requires firms to recognize on the balance sheet the assets and liabilities arising from lease contracts (FASB, 2016).

We focus on these two standards for three reasons. First, both ASC 606 and ASC 842 are among the most complex accounting standards introduced in the past decade, and their implementation likely required substantial overhauls of firms' financial reporting systems. The economic significance of their effect is reflected by the preponderance of recent prior studies that use ASC 606 and/or ASC 842 to examine how accounting standards affect internal information systems. Second, both standards were announced well in advance of their effective dates, providing firms with a clear window to invest in and improve internal information systems. Third, this shock-based research design aligns with a growing literature that uses regulatory changes in accounting standards to study the quality of firms' internal information environments.[25] In particular, prior studies suggest that changes in accounting standards affect firms' internal information environments outside the targeted scope of the specific standard.

To implement a test, we apply a non-staggered continuous difference-in-differences design layered on top of Eq. (2) (Atanasov and Black, 2016, 253–254). Although the announcements occur at the same time for every firm (i.e., the shocks are non-staggered), the ultimate impact of the changes in accounting standards likely varies in magnitude across firms (i.e., the treatment is continuous), namely in proportion with reliance on the accounting standard. For example, the impact of ASC 606 and ASC 842 is likely small if a firm does not have complex contracts with customers and has no operating lease arrangements, respectively. Therefore, for each shock, we estimate the degree to which firms are affected by the change in accounting standards (i.e., their treatment intensity). Because firms may also make operational changes in response to these announcements, we compute a static firm-level treatment intensity measure over the pre-event period. The event window spans four years before and after the announcement of each standard, allowing us to compare pre- and post-period trends across firms with different *ex ante* levels of exposure.

For ASU 2015-14 (i.e., the deferral of ASU, 2014-09, Revenue from Contracts with Customers), we measure treatment intensity as the number of unique customers disclosed in the firm's 10-K filing. The rationale is that ASC 606 has a greater impact on firms with complex customer contracts involving enforceable rights and obligations, particularly when those firms maintain relationships with multiple large customers.[26] To capture this, we rely on firms' mandated disclosures under SEC Regulation S-K, which require the identification of customers accounting for more than 10% of consolidated sales revenue. For ASU 2016-02 (i.e., Leases), we define treatment intensity as the total capitalized amount of operating lease obligations, reflecting the extent to which a firm relies on operating lease arrangements. To account for differences in firm scale, we normalize each treatment intensity measure by firm size. In both cases, we restrict the sample to firms with complete data across the full event window to ensure consistency in estimating treatment effects.

To address concerns about non-parallel trends and further align with the guidance of Atanasov and Black (2016, 258–259), we augment our continuous difference-in-differences design by incorporating entropy balancing.[27] This approach ensures that observable covariates are well balanced between relatively more treated and less treated firms. Specifically, for each shock, we split firms into two groups based on the median value of treatment intensity. We then implement entropy balancing to match the first moment of the covariates used in Eq. (2) between the two groups, using data from the pre-event period. These weights are held fixed throughout the analysis, producing covariate balance across treated and control firms. After balancing, all observable differences between the two groups are statistically insignificant.

Using this setup, we perform two analyses for each accounting standard announcement. First, we directly examine changes in IIQ around the announcements to establish a basis for examining whether and how incentive contract design changes following improvements in IIQ. Second, we examine changes in the details of CEOs' incentive contracts around the announcements. We tabulate results from these tests in Tables 7 and 8, with the two tables presenting, respectively, results for ASU 2015-14 and ASU 2016-02.

Regarding the first analysis, Table 7 Panel A shows that the coefficient on *Post* × *Treatment* is positive and statistically significant. This finding indicates that firms affected by ASU 2015-14 see improvements in their internal information environment. We find similar

[24] Anecdotal evidence exists such as that from Autodesk, Inc., which noted that, likely in response to ASC 606, the firm "adopted new performance metrics for their bonus and equity plans that align with the key drivers of success during the business model transition and reflect the health of the business during the transition." See, for instance, page 27 in their first proxy statement after the effective date of the ASU 2014-09 (Autodesk Inc, 2017). This example highlights how regulatory-induced improvements in internal reporting systems can lead to substantive changes in performance metric design.

[25] See, for instance, Gallemore and Labro (2015), Shroff (2017), Cheng et al. (2018), Heitzman and Huang (2019), Gelsomin (2022), Chatterjee (2023), Kim et al. (2025), Christensen et al. (2025a, 2025b), and Roh (2025).

[26] This treatment intensity is intended to capture a particular type of exposure: the presence of concentrated, large-scale contractual relationships (that are observable). Firms with one or more major customers are more likely to engage in bespoke contractual arrangements, including customized performance obligations, negotiated delivery schedules, and tailored pricing structures. Under the new regulation, such firms are more likely to face non-trivial implementation challenges, as they must reassess and disaggregate revenue across these individualized contracts, often requiring significant changes to internal systems. In a similar vein, Ali and Tseng (2025) find that ASC 606 had a greater impact on firms that have long revenue generating cycles, such as those using the percentage-of-completion method, which likely have larger customers. However, we recognize that this treatment intensity may not fully capture the breadth of firm-level exposure to ASC 606 as there may be multiple dimensions of treatment intensity relevant to the revenue recognition standard (e.g., firms with many smaller customers may face implementation challenges because of variation in bundled offerings across customers).

[27] In unreported analyses, we find similar inferences without incorporating entropy balancing, suggesting our results are not sensitive to this adjustment.

results for ASU 2016-02 in Table 8 Panel A. Consistent with prior literature, these findings suggest that firms' internal information environment improve if they are affected by changes in accounting standards.

Regarding the effect of these changes on CEOs' incentive contracts, we first examine our main proxies—*Number of Metrics* and *Metrics Dissimilarity*—alongside their disaggregated financial and non-financial components. Results for ASU 2015-14 and ASU 2016-02 are in Tables 7 and 8 Panels B and C, respectively. Broadly, we find that the coefficient on *Post* × *Treatment* is positive and statistically significant in specifications focused on financial performance metrics. These findings are consistent with the notion that, as firms upgrade their internal reporting systems in preparation for the new accounting standards, they also learn about their operations and value chains and/or identify new performance dimensions to incorporate into executive incentive contracts, mainly in financial aspects given the financial focus on the accounting standards. One exception is ASU 2015-14, for which we also find evidence that firms increase the dissimilarity of their contracts along non-financial dimensions relative to their industry peers. This suggests that the internal reporting upgrades prompted by the accounting standard may have also enhanced firms' ability to define or track non-financial performance dimensions, enabling greater customization in incentive design beyond purely financial metrics.

To provide further insight, we examine firms' use of specific performance metric categories in greater detail. Specifically, in Tables 7 and 8 Panel D, we examine changes in the inclusion of various performance metric types following the accounting standards. Two patterns emerge from the data. First, consistent with the notion that both standards pertain to core accounting topics, affected firms increase their use of accounting-based performance metrics. Second, we find suggestive evidence that these additions are offset by a decline in the use of non-GAAP performance metrics, indicating a shift in emphasis toward standardized accounting metrics, which improved in quality because of the new accounting standards. We also find that firms affected by ASU 2015-14 are more likely to implement revenue-based metrics, whereas those affected by ASU 2016-02 are more likely to introduce EPS-based performance metrics, largely consistent with the focus of the respective accounting standard.

Overall, the shock-based analyses support the interpretation that IIQ acts as a constraint on performance metric design. Firms more affected by accounting standards improve their internal information environments and adjust incentive contracts—particularly by increasing financial metrics aligned with the nature of the reform. These patterns suggest that when firms are compelled to upgrade reporting infrastructure, the marginal cost of implementing tailored metrics declines, enabling boards to design more customized incentive contracts.[28]

6. Additional empirical tests and results

Thus far, the findings suggest that IIQ acts as a limiting factor in the selection of performance metrics for CEO incentive contracts. This evidence raises two additional questions, which we examine in this section.[29] First, in which types of performance metrics does IIQ play a larger role? And second, what potential impact does higher IIQ have, if any?

6.1. In which types of performance metrics does IIQ play a role?

Our main hypothesis is that improved IIQ reduces processing costs associated with performance metrics. This reduction, in turn, allows boards to incorporate a broader and more dissimilar set of performance metrics into executive incentive contracts. In this subsection, we examine whether IIQ has a different association with financial or non-financial performance metrics. Our composite proxy, which is based on both financial and non-financial components, is designed to capture the overall quality of a firm's internal information environment. Therefore, it is conceptually possible and theoretically plausible that IIQ is more critical for non-financial performance metrics, which may lack long-standing institutional familiarity among executives, directors, and investors. Conversely, the association may be stronger for financial metrics, given their higher reporting frequency and the sustained monitoring focus they receive from boards, analysts, and investors. We therefore approach the analysis empirically, seeking to shed light on the relative role of IIQ in supporting different types of performance metrics.

To do so, we re-estimate Eq. (2) for our main proxies—*Number of Metrics* and *Metrics Dissimilarity*—alongside their disaggregated financial and non-financial components. We use both our composite IIQ measure and its seven individual proxies (see Section 3.3 for details). This approach also allows us to assess construct validity and ensure that our results are not specific to any one measure but instead reflect the broader theoretical construct.

Results are in Table 9 Panel A, with each cell presenting results from one estimation. Across specifications focused on non-financial

[28] We caveat that the shock-based analyses also are not purely about the supply side of performance metric choices. Through implementing new accounting standards, firms may learn more about their operations and value chains, driving the demand side of performance metric choices as well. Therefore, it is possible that our shock-based analyses partially capture an increase in demand of performance metrics, which triggers an investment in IIQ, which then results in more efficient usage of performance metrics in incentive contracts.

[29] In the Online Appendix (Table OA2), we also examine whether the relation between IIQ and performance metric selection is driven by correlated omitted variables by examining strategic shifts coinciding with external CEO appointments, which we find to have no material impact on the relation between IIQ and performance metric selection.

Table 7
Internal information quality and performance metric selection—ASU 2015-14.

Panel A. Changes in internal information quality around ASU 2015-14

Variable	(1)
	Dependent variable:
	IIQ$_t$
Post × *Treatment*	0.466***
	(0.129)
Controls	yes
Fixed effects	firm, year
Observations	4,238
Adjusted R^2	86.936%

Panel B. Changes in the number of performance metrics around ASU 2015-14

Variable	Dependent variable:		
	(1)	(2)	(3)
	Number of Metrics$_{t+1}$	*Number of Financial Metrics*$_{t+1}$	*Number of Non-Financial Metrics*$_{t+1}$
Post × *Treatment*	0.700	1.309***	−0.612
	(0.641)	(0.368)	(0.425)
Controls	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year
Observations	4,238	4,238	4,238
Adjusted R^2	57.505%	54.561%	59.984%

Panel C. Changes in the dissimilarity of performance metrics around ASU 2015-14

Variable	(1)	(2)	(3)
	Dependent variable:		
	Metrics Dissimilarity$_{t+1}$	*Financial Metrics Dissimilarity*$_{t+1}$	*Non-Financial Metrics Dissimilarity*$_{t+1}$
Post × *Treatment*	1.361***	1.604***	3.237***
	(0.481)	(0.344)	(0.363)
Controls	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year
Observations	4,238	4,238	4,238
Adjusted R^2	47.861%	41.336%	49.636%

Panel D. Changes in the use of specific performance metric categories around ASU 2015-14

Variable	(1)	(2)	(3)
	Dependent variable:		
	Number of Accounting Metrics$_{t+1}$	*Number of Non-GAAP Metrics*$_{t+1}$	*Number of Revenue Metrics*$_{t+1}$
Post × *Treatment*	1.591***	−0.175**	1.879***
	(0.412)	(0.073)	(0.143)
Controls	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year
Observations	4,238	4,238	4,238
Adjusted R^2	54.221%	35.765%	69.425%

This table presents results from a non-staggered, continuous difference-in-differences design estimating the effects of ASU 2015-14 (i.e., deferral of ASU, 2014-09 "Revenue from Contracts with Customers") on firms' internal information quality and performance metric selection. Panels A through D present, respectively, results for internal information quality, the number of performance metrics, the dissimilarity of the focal CEO's incentive contract relative to industry peers, and changes in the use of specific performance metric categories following the accounting standard. For Panels B and C, we present results both pooled and separately for financial and non-financial performance metrics. The estimation window spans four years before and after each event. We construct a static firm-level treatment intensity measure and entropy balance firms based on all control variables using data from the pre-event period. *Post* is an indicator variable equal to one for fiscal periods beginning after the announcement date of ASU 2015-14 on July 9, 2015, and zero otherwise. *Treatment* is the number of unique customers disclosed in the firm's 10-K, scaled by firm size. The regression models also include firm and year effects and *Delta*, *Vega*, *Age*, *Tenure*, *Number of Rivals*, *Rival Similarity*, *Sales*, *Book-to-Market*, *Leverage*, *Sales Growth*, *Cash*, *CEO Duality*, *Board Size*, *Board Independence*, and *Board Experience*; for brevity, we do not report their coefficient estimates and standard errors. Standard errors are in parentheses and are adjusted for within-cluster correlation at the level of treatment (i.e., firm). *, **, and *** denote that the coefficient is significantly different from zero at, respectively, two-tailed probability levels of 10%, 5%, and 1%. Appendix A defines all variables.

performance metrics, most IIQ proxies load positively and significantly, suggesting that information quality is particularly useful for implementing non-financial metrics.[30] These findings are consistent with the idea that non-financial metrics may be less standardized or familiar and thus overall require greater informational infrastructure.[31] The finding that individual proxies that have a financial nature (e.g., forecast accuracy) exhibit results also in the non-financial aspect likely suggests that the proxy captures a broader improvement in firms' overall internal information system that also has a non-financial dimension. Alternatively, the proxy may not directly capture non-financial dimensions itself. However, better understanding of future financial performance may still have spill-over effects to non-financial parts of the system (Christensen et al., 2025b), for example, through better assessment of the financial impact of non-financial investments (Shroff, 2017; Cheng et al., 2018). One exception to this pattern is *EA Speed*, which is strongly associated with financial performance metrics. This result suggests that the timeliness of earnings disclosure may reflect internal processes particularly relevant for tracking financial outcomes.

We also exploit this split between financial and non-financial metrics in our cross-sectional analyses to further corroborate the validity of our inferences. In particular, external environmental and social pressure have a clearer link with non-financial metrics compared to other cross-sectional variables we examine in Section 4.2. If external environmental and social pressure indeed push boards to adopt a broader and more tailored set of performance metrics under higher IIQ, it is natural to predict that such results should be stronger for non-financial metrics. To test this conjecture, we repeat our analysis in Table 6 separately for financial and non-financial metrics in Table 9 Panels B and C. Across both panels, we find that the positive coefficient of *IIQ* is significantly larger for the subsample with high *E-Risk* or *S-Risk* when we examine non-financial metrics, but we do not observe a similar difference for financial metrics. This finding is consistent with the non-financial nature of the agency conflicts captured by the cross-sectional variables *E-Risk* and *S-Risk*.

6.2. Does IIQ mitigate negative performance effects of complex compensation structures?

Our findings with regard to the role of IIQ suggest that firms with higher IIQ may be better positioned to use broader and more tailored compensation structures, which are typically more complex. In contrast, prior research suggests that complex compensation structures adopted for other reasons (e.g., external pressures from compensation consultants and proxy advisors) can be associated with weaker performance outcomes (Albuquerque et al., 2025). We therefore examine whether IIQ is associated with differences in the relation between complex compensation structures and future performance, particularly when a broader and more customized set of performance metrics is used. To test this idea, we examine whether the breadth and dissimilarity of performance metrics are associated with future profitability as well as whether these associations depend on IIQ. We therefore estimate the relation between future profitability, IIQ, and both the number and dissimilarity of financial and non-financial performance metrics using the following specification:

$$[ROA_{it+n}] = \beta_1 \bullet (IIQ_{it} \times [Financial\ Metrics_{it}]) + \beta_2 \bullet (IIQ_{it} \times [Non\text{-}Financial\ Metrics_{it}]) + \beta_3 \bullet [Financial\ Metrics_{it}] + \beta_4 \bullet [Non\text{-}Financial\ Metrics_{it}] + \beta_5 \bullet IIQ_{it} + \Phi \bullet X_{it} + \mathrm{T} \bullet \tau_i + \Omega \bullet \xi_t + \varepsilon_{it+n}, \tag{3}$$

where $n \in \{1, 2, 3\}$, $[ROA_{it+n}]$ is firm *i*'s return on assets in percentage points from year *t* to $t+n$, [*Financial Metrics*] is either *Number of Financial Metrics* or *Financial Metrics Dissimilarity*, and [*Non-Financial Metrics*] is either *Number of Non-Financial Metrics* or *Non-Financial Metrics Dissimilarity*. To ease the interpretation of the coefficient estimates, we use decile ranks of *IIQ*, [*Financial Metrics*], and [*Non-Financial Metrics*] scaled to range from zero to one. As a result, β_1 and β_2 capture how the impact of the highest levels of contract complexity on profitability changes when moving from firm-years with the lowest IIQ to firm-years with the highest levels of IIQ.

Table 10 presents the results for one-, two-, and three-year ahead profitability, with the two panels presenting, respectively, results for the number of performance metrics and the dissimilarity of performance metrics relative to industry peers. Across both panels, we find that non-financial metrics—when used in conjunction with high IIQ—are positively associated with future profitability, whereas their standalone use is negatively associated with future profitability. These findings suggest a role of IIQ as an important input for incentive contracts that incorporate more numerous and more dissimilar non-financial metrics. We emphasize, however, that these results are correlational and should not be interpreted as causal evidence that higher IIQ leads to improved decision-making through the design of better performance metrics. Our results do not allow us to disentangle this explanation from one in which both the design of incentive contracts and future performance are jointly related to firms' current IIQ.

[30] For forecast accuracy, we code the variable as zero if there is no management guidance. In the Online Appendix (Table OA3), we also examine alternative treatments for observations without management guidance. We find that our results are robust to such alternative treatments. We also examine whether forecast provision alone is associated with contract outcomes, by including an indicator variable for EPS forecast issuance. The indicator is not significantly associated with any of the contract outcome variables.

[31] The cross-sectional test based on comovement of performance metrics in Section 4.2.2 may seem to suggest that IIQ is of higher importance when metrics are easier to process. However, that cross-sectional test examines *when IIQ can matter* rather than *how IIQ matters* as in Table 9. If boards have sufficient processing capacity because it is easier to examine multiple metrics, then IIQ is the more binding constraint and becomes more relevant. However, that cross-sectional test is agnostic about which metrics become sensitive to IIQ once IIQ becomes more relevant. For example, if boards can process and understand financial metrics in a more efficient manner, it may have more capacity to spend on examining non-financial metrics to include in the contract. If so, IIQ with regard to such non-financial metrics will matter more for their inclusion in the contract. In contrast, if boards have difficulty processing financial metrics, it may not be able to allocate sufficient capacity to examine non-financial metrics, thus making the IIQ with regard to such non-financial metrics less relevant with regard to the final design of the contract.

Table 8
Internal information quality and performance metric selection—ASU 2016-02.

Panel A. Changes in internal information quality around ASU 2016-02

Variable	(1)
	Dependent variable:
	IIQ_t
Post × *Treatment*	0.043*
	(0.025)
Controls	yes
Fixed effects	firm, year
Observations	4,228
Adjusted R^2	86.160%

Panel B. Changes in the number of performance metrics around ASU 2016-02

Variable	(1)	(2)	(3)
	Dependent variable:		
	Number of Metrics$_{t+1}$	*Number of Financial Metrics*$_{t+1}$	*Number of Non-Financial Metrics*$_{t+1}$
Post × *Treatment*	0.224	0.184***	0.039
	(0.154)	(0.069)	(0.125)
Controls	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year
Observations	4,228	4,228	4,228
Adjusted R^2	55.586%	54.615%	53.966%

Panel C. Changes in the dissimilarity of performance metrics around ASU 2016-02

Variable	(1)	(2)	(3)
	Dependent variable:		
	Metrics Dissimilarity$_{t+1}$	*Financial Metrics Dissimilarity*$_{t+1}$	*Non-Financial Metrics Dissimilarity*$_{t+1}$
Post × *Treatment*	0.120	0.198**	0.070
	(0.126)	(0.080)	(0.113)
Controls	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year
Observations	4,228	4,228	4,228
Adjusted R^2	46.150%	42.540%	44.981%

Panel D. Changes in the use of specific performance metric categories around ASU 2016-02

Variable	(1)	(2)	(3)
	Dependent variable:		
	Number of Accounting Metrics$_{t+1}$	*Number of Non-GAAP Metrics*$_{t+1}$	*Number of EPS Metrics*$_{t+1}$
Post × *Treatment*	0.254***	−0.028**	0.068**
	(0.081)	(0.013)	(0.027)
Controls	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year
Observations	4,228	4,228	4,228
Adjusted R^2	56.541%	34.148%	58.346%

This table presents results from a non-staggered, continuous difference-in-differences design estimating the effects of ASU 2016-02 ("Leases") on firms' internal information quality and performance metric selection. Panels A through D present, respectively, results for internal information quality, the number of performance metrics, the dissimilarity of the focal CEO's incentive contract relative to industry peers, and changes in the use of specific performance metric categories following the accounting standard. For Panels B and C, we present results both pooled and separately for financial and non-financial performance metrics. The estimation window spans four years before and after each event. We construct a static firm-level treatment intensity measure and entropy balance firms based on all control variables using data from the pre-event period. *Post* is an indicator variable equal to one for fiscal periods beginning after the announcement date of ASU 2016-02 on February 25, 2016, and zero otherwise. *Treatment* is the total capitalized amount of operating lease obligations, scaled by firm size. The regression models also include firm and year effects and *Delta*, *Vega*, *Age*, *Tenure*, *Number of Rivals*, *Rival Similarity*, *Sales*, *Book-to-Market*, *Leverage*, *Sales Growth*, *Cash*, *CEO Duality*, *Board Size*, *Board Independence*, and *Board Experience*; for brevity, we do not report their coefficient estimates and standard errors. Standard errors are in parentheses and are adjusted for within-cluster correlation at the level of treatment (i.e., firm). *, **, and *** denote that the coefficient is significantly different from zero at, respectively, two-tailed probability levels of 10%, 5%, and 1%. Appendix A defines all variables.

Table 9
Internal information quality and performance metric selection—financial versus non-financial performance metrics.

Panel A. Individual proxies for internal information quality

Variable	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable:			Dependent variable:		
	Number of Metrics$_{t+1}$	*Number of Financial Metrics*$_{t+1}$	*Number of Non-Financial Metrics*$_{t+1}$	*Metrics Dissimilarity*$_{t+1}$	*Financial Metrics Dissimilarity*$_{t+1}$	*Non-Financial Metrics Dissimilarity*$_{t+1}$
IIQ	0.192***	0.027	0.166***	0.158***	0.035	0.153***
	(0.074)	(0.047)	(0.051)	(0.060)	(0.037)	(0.041)
By proxy estimations (each cell refers to one estimation):						
Forecast Accuracy	0.301***	0.103**	0.197***	0.147***	0.044	0.119***
	(0.063)	(0.044)	(0.044)	(0.051)	(0.035)	(0.037)
EA Speed	1.696	4.476**	−2.780	3.752	5.031***	−1.310
	(3.533)	(1.815)	(3.057)	(2.815)	(1.495)	(2.404)
Audit Simplicity	0.710*	−0.137	0.846***	0.300	−0.134	0.583***
	(0.398)	(0.234)	(0.284)	(0.302)	(0.189)	(0.214)
No SOX404 Weakness	0.409*	0.105	0.304**	0.221	0.136	0.233**
	(0.212)	(0.145)	(0.139)	(0.179)	(0.121)	(0.107)
No Restatement	0.230	−0.042	0.271	0.214	−0.023	0.214
	(0.224)	(0.139)	(0.169)	(0.184)	(0.112)	(0.138)
Environmental Initiatives	0.378*	−0.088	0.467**	0.262	−0.107	0.427***
	(0.205)	(0.117)	(0.182)	(0.166)	(0.091)	(0.142)
ESG Reporting Scope	0.254*	0.112	0.142*	0.184	0.073	0.127*
	(0.130)	(0.078)	(0.082)	(0.115)	(0.067)	(0.072)
Controls	yes	yes	yes	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year	firm, year	firm, year	firm, year
Observations	20,409	20,409	20,409	20,409	20,409	20,409
Average adjusted R^2	50.400%	48.887%	49.799%	45.542%	41.770%	43.688%

Panel B. Environmental reputation risk—financial versus non-financial performance metrics

Subsample—split by *E-Risk*	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	low	high	low	high	low	high	low	high
Variable	Dependent variable:		Dependent variable:		Dependent variable:		Dependent variable:	
	Number of Financial Metrics$_{t+1}$		*Number of Non-Financial Metrics*$_{t+1}$		*Financial Metrics Dissimilarity*$_{t+1}$		*Non-Financial Metrics Dissimilarity*$_{t+1}$	
IIQ	0.009	0.054	0.085	0.278**	0.058	0.069	0.103**	0.272**
	(0.076)	(0.093)	(0.059)	(0.134)	(0.062)	(0.074)	(0.051)	(0.106)
Controls	yes	yes	yes	yes	yes	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year	firm, year	firm, year	firm, year	firm, year	firm, year
Observations	9,140	4,257	9,140	4,257	9,140	4,257	9,140	4,257
Adjusted R^2	46.609%	47.564%	0.307%	0.513%	38.351%	43.996%	41.597%	38.100%

(*continued on next page*)

Table 9 (*continued*)

Panel B. Environmental reputation risk—financial versus non-financial performance metrics								
Subsample—split by *E-Risk*	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	low	high	low	high	low	high	low	high
Variable	Dependent variable:		Dependent variable:		Dependent variable:		Dependent variable:	
	Number of Financial Metrics$_{t+1}$		*Number of Non-Financial Metrics*$_{t+1}$		*Financial Metrics Dissimilarity*$_{t+1}$		*Non-Financial Metrics Dissimilarity*$_{t+1}$	
Difference in *IIQ*	0.045		0.193*					
					0.010		0.169*	

Panel C. Social reputation risk—financial versus non-financial performance metrics								
Subsample—split by *S-Risk*	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	low	high	low	high	low	high	low	high
Variable	Dependent variable:		Dependent variable:		Dependent variable:		Dependent variable:	
	Number of Financial Metrics$_{t+1}$		*Number of Non-Financial Metrics*$_{t+1}$		*Financial Metrics Dissimilarity*$_{t+1}$		*Non-Financial Metrics Dissimilarity*$_{t+1}$	
IIQ	−0.059	0.008	0.085	0.283***	−0.012	0.053	0.111	0.276***
	(0.092)	(0.082)	(0.079)	(0.107)	(0.078)	(0.065)	(0.068)	(0.085)
Controls	yes	yes	yes	yes	yes	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year	firm, year	firm, year	firm, year	firm, year	firm, year
Observations	7,182	6,215	7,182	6,215	7,182	6,215	7,182	6,215
Adjusted R^2	46.376%	48.452%	50.154%	46.290%	37.281%	43.530%	42.999%	38.843%
Difference in *IIQ*	0.066		0.198*					
					0.065		0.165*	

This table presents results from examining financial versus non-financial performance metrics. Panel A presents results from examining the robustness of the relation between internal information quality and details of CEOs' incentive contracts, using all proxies for internal information quality. Panels B and C present, respectively, results from repeating the analysis in Table 6, which examines cross-sectional variation based on external environmental and social pressure. The regression models also include firm and year effects and *Delta*, *Vega*, *Age*, *Tenure*, *Number of Rivals*, *Rival Similarity*, *Sales*, *Book-to-Market*, *Leverage*, *Sales Growth*, *Cash*, *CEO Duality*, *Board Size*, *Board Independence*, and *Board Experience*; for brevity, we do not report their coefficient estimates and standard errors. Standard errors are in parentheses and are adjusted for within-cluster correlation at the level of treatment (i.e., firm). *, **, and *** denote that the coefficient is significantly different from zero at, respectively, two-tailed probability levels of 10%, 5%, and 1%. Appendix A defines all variables.

Table 10
Internal information quality, performance metric selection, and future profitability.

Panel A. Number of performance metrics and future profitability

Variable	(1)	(2)	(3)
	Dependent variable:		
	ROA_{t+1}	ROA_{t+2}	ROA_{t+3}
Number of Non-Financial Metrics	−2.560***	−2.368***	−2.002***
	(0.598)	(0.623)	(0.565)
IIQ × *Number of Non-Financial Metrics*	2.727***	2.461***	2.039**
	(0.856)	(0.930)	(0.877)
Number of Financial Metrics	0.820	0.429	0.294
	(0.531)	(0.677)	(0.645)
IIQ × *Number of Financial Metrics*	−0.868	−0.576	−0.297
	(0.860)	(1.013)	(0.956)
IIQ	−0.569	−1.017	−0.974
	(0.691)	(0.830)	(0.787)
Controls	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year
Observations	19,915	19,915	19,915
Adjusted R^2	51.679%	57.689%	62.885%

Panel B. Dissimilarity of performance metrics and future profitability

Variable	(1)	(2)	(3)
	Dependent variable:		
	ROA_{t+1}	ROA_{t+2}	ROA_{t+3}
Non-Financial Metrics Dissimilarity	−0.162***	−0.188***	−0.170***
	(0.062)	(0.066)	(0.061)
IIQ × *Non-Financial Metrics Dissimilarity*	0.206**	0.254***	0.234***
	(0.084)	(0.091)	(0.084)
Financial Metrics Dissimilarity	0.225	0.349	0.415
	(0.501)	(0.606)	(0.559)
IIQ × *Financial Metrics Dissimilarity*	−0.025	−0.217	−0.284
	(0.782)	(0.917)	(0.855)
IIQ	−0.722	−1.299	−1.182
	(0.649)	(0.795)	(0.747)
Controls	yes	yes	yes
Fixed effects	firm, year	firm, year	firm, year
Observations	19,915	19,915	19,915
Adjusted R^2	51.606%	57.659%	62.869%

This table presents results from examining whether the relation between details of CEOs' incentive contracts and firms' future profitability varies with internal information quality. Panels A and B present, respectively, results for the number of performance metrics and the dissimilarity of the focal CEO's incentive contract relative to industry peers. The regression models also include firm and year effects and *Delta*, *Vega*, *Age*, *Tenure*, *Number of Rivals*, *Rival Similarity*, *Sales*, *Book-to-Market*, *Leverage*, *Sales Growth*, *Cash*, *CEO Duality*, *Board Size*, *Board Independence*, and *Board Experience*; for brevity, we do not report their coefficient estimates and standard errors. Standard errors are in parentheses and are adjusted for within-cluster correlation at the level of treatment (i.e., firm). *, **, and *** denote that the coefficient is significantly different from zero at, respectively, two-tailed probability levels of 10%, 5%, and 1%. Appendix A defines all variables.

7. Summary and conclusion

In this study, we examine the relation between firms' IIQ and performance metric selection for CEO incentive contracts. Specifically, we test the hypothesis that the quality of a firm's internal information is a friction in boards' performance metric selection. Our main finding is that boards enlarge the set of performance metrics and also incorporate metrics that are more dissimilar to what industry peers use when firms' IIQ is higher. We also find that this positive relation between IIQ and the breadth and variety of performance metrics is stronger in settings with differing demands for tailored performance metrics: (1) firms with greater organizational complexity; (2) firms for which internal information is easier to process and act upon; and (3) firms facing heightened external environmental and social pressures.

Collectively, these findings are consistent with our hypothesis that IIQ is a binding friction in boards' performance metric selection. Results from two shock-based analyses in which we examine changes in IIQ that are likely induced by plausibly exogenous shifts in two financial accounting standards further support this interpretation. In a similar vein, we show that a broader and more dissimilar set of metrics—particularly non-financial metrics—is positively associated with future profitability when used in conjunction with high IIQ.

Our study's findings are relevant in light of the growing prevalence of novel non-financial performance metrics and performance-vested equity grants. A key insight from our study is that internal information systems can serve as a strategic lever, enabling boards to construct more effective, context-specific incentive contracts. Future research can build on these insights by identifying which types of information system improvements are most effective for supporting tailored CEO incentive contracts.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jacceco.2026.101894.

Appendix A. Variable definitions

Variable	Description
Number of Metrics	The number of performance metrics in the CEO's incentive contract. Data come from ISS Incentive Lab.
Number of Financial Metrics	The number of financial performance metrics in the CEO's incentive contract. Financial performance metrics are those in the following categories: (1) accounting; (2) activity-related; (3) balance sheet-related; (4) cash flow; (5) earnings-related; (6) economic value; (7) investment ratios; (8) market-related; (9) revenue-related; (10) solvency-related; and (11) stock price. Data come from ISS Incentive Lab.
Number of Non-Financial Metrics	The number of non-financial performance metrics in the CEO's incentive contract. Non-financial performance metrics are those in the following categories: (1) CSR; (2) environmental; (3) non-financial; (4) social; and (5) other. Data come from ISS Incentive Lab.
Metrics Dissimilarity	The dissimilarity of the focal CEO's incentive contract relative to industry peer CEOs' incentive contracts. For each industry-year defined using the Fama and French (1997) 48 industry groups, we first calculate the average number of performance metrics used in each of 16 performance categories across all firms. We then compute contract dissimilarity as the sum of absolute differences between the focal CEO's contract and the industry-year average, taken across all 16 categories. Data come from ISS Incentive Lab and Compustat.
Financial Metrics Dissimilarity	The dissimilarity of the focal CEO's incentive contract relative to industry peer CEOs' incentive contracts, calculated only using the 11 financial performance categories.
Non-Financial Metrics Dissimilarity	The dissimilarity of the focal CEO's incentive contract relative to industry peer CEOs' incentive contracts, calculated only using the 5 non-financial performance categories.
Number of Accounting Metrics	The number of accounting-based performance metrics in the CEO's incentive contract. Data come from ISS Incentive Lab.
Number of Non-GAAP Metrics	The number of non-GAAP-based performance metrics in the CEO's incentive contract. Data come from ISS Incentive Lab.
Number of Revenue Metrics	The number of revenue-based performance metrics in the CEO's incentive contract. Data come from ISS Incentive Lab.
Number of EPS Metrics	The number of EPS-based performance metrics in the CEO's incentive contract. Data come from ISS Incentive Lab.
IIQ	Internal information quality index, computed as the first principal component of the following variables: *Forecast Accuracy*, *EA Speed*, *Audit Simplicity*, *No SOX404 Weakness*, *No Restatement*, *Environmental Initiatives*, and *ESG Reporting Scope*. This variable explains 24.906% of the total variance across all seven measures, with its eigenvalue exceeding 1 at $0.249 \times 7 \approx 1.743$. See data sources below.
Forecast Accuracy	Average of the absolute difference between management's quarterly estimate of earnings per share and the firm's realized quarterly earnings per share, computed over the past twelve quarters. This variable is multiplied by negative one, such that larger values correspond to higher internal information quality (Gallemore and Labro, 2015). To retain as many observations as possible, we set *Forecast Accuracy* to zero when the firm does not provide a forecast about earnings. Data come from I/B/E/S.
EA Speed	Number of days between the end of the fiscal year and the firm's earnings announcement (Gallemore and Labro, 2015). To ease the interpretation of this variable, we divide it by 365 and multiply it by negative one such that larger values correspond to faster earnings announcements. Data come from Compustat.
Audit Simplicity	Total audit fees scaled by annual revenue. To ease the interpretation of this variable, we multiply it by negative one such that larger values correspond to less audit-intensive firms. Data come from Audit Analytics Audit Fees.
No SOX404 Weakness	An indicator equal to one if the firm did not report a Section 404 material weakness in any of the past three years, zero otherwise (Gallemore and Labro, 2015). Data come from Audit Analytics SOX404 Internal Controls.
No Restatement	An indicator equal to one if the firm had no restatement in any of the past three years, zero otherwise, zero otherwise (Gallemore and Labro, 2015). Data come from Audit Analytics Financial Restatements.
Environmental Initiatives	An indicator equal to one if the firm reports making proactive environmental investments or expenditures to reduce future risks or increase future opportunities, zero otherwise. Data come from LSEG ESG (formerly known as Refinitiv and ASSET4).
ESG Reporting Scope	An indicator equal to one if the firm's environmental and social reporting covers all of its global activities, based on (1) number of employees, (2) group revenue, or (3) number of group operations, zero otherwise. Data come from LSEG ESG (formerly known as Refinitiv and ASSET4).
Delta	Sensitivity of the risk-neutral value of the CEO's portfolio of stock and stock options to a 1% change in the price of the underlying stock (Core and Guay, 2002). Because this variable is skewed, we use the natural logarithm of one plus the raw value in our analyses. Data come from ExecuComp and CRSP.
Vega	Sensitivity of the risk-neutral value of the CEO's portfolio of stock options to a 0.01 change in the volatility of the underlying stock (Core and Guay, 2002). Because this variable is skewed, we use the natural logarithm of one plus the raw value in our analyses. Data come from ExecuComp and CRSP.
Age	The CEO's age. Because this variable is skewed, we use the natural logarithm of one plus the raw value in our analyses. Data come from ExecuComp.
Tenure	Number of years the CEO has held his/her office. Because this variable is skewed, we use the natural logarithm of one plus the raw value in our analyses. Data come from ExecuComp.

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Variable	Description
Number of Rivals	Number of product market rivals, as identified by Hoberg and Phillips (2010, 2016). Because this variable is skewed, we use the natural logarithm of one plus the raw value in our analyses. Data come from Hoberg-Phillips Data Library.
Rival Similarity	Median similarity to the firm's product market rivals, as identified by Hoberg and Phillips (2010, 2016). Data come from Hoberg-Phillips Data Library.
Sales	Annual revenue. Because this variable is skewed, we use the natural logarithm of one plus the raw value in our analyses. Data come from Compustat.
Book-to-Market	Ratio of book value of total assets to the firm's market value. Data come from Compustat.
Leverage	Book value of total long-term debt, scaled by total assets. Data come from Compustat.
Sales Growth	Growth in annual revenue over the prior year. Data come from Compustat.
Cash	Cash and cash equivalents balance, scaled by total assets. Data come from Compustat.
CEO Duality	An indicator equal to one if the CEO is also the Chairman of the Board, zero otherwise. Data come from BoardEx.
Board Size	Number of board members. Because this variable is skewed, we use the natural logarithm of one plus the raw value in our analyses. Data come from BoardEx.
Board Independence	Fraction of board members who are independent. Data come from BoardEx.
Board Experience	Fraction of board members who sit on three or more other boards. Data come from BoardEx.
Segments	Number of business segments. Data come from Compustat Segments.
σReturns	Historical idiosyncratic stock return volatility, defined as the standard deviation of the firm's annualized returns that are unexplained by the Carhart (1997) four-factor model estimated over the prior 60 months. Data come from CRSP and Fama-French Portfolios and Factors.
Comovement	Correlation between quarterly stock returns and earnings per share estimated over a 20-quarter period. Data come from Compustat Quarterly.
E-Risk	Environmental reputation risk. Data come from RepRisk and are available until 2020.
S-Risk	Social reputation risk. Data come from RepRisk and are available until 2020.
ROA	Income scaled by total assets over the period *t* until the subscripted period.

This table presents variable definitions.

References

Abadie, A., Athey, S., Imbens, G., Wooldridge, J.M., 2023. When should you adjust standard errors for clustering? Q. J. Econ. 138 (1), 1–35.

Abbott Laboratories, 2024. Proxy Statement 2023. https://www.sec.gov/ix?doc=/Archives/edgar/data/0000001800/000130817924000182/abt4257331-def14a.htm.

Albuquerque, A.M., Carter, M.E., Guo, Z., Lynch, L.J., 2025. Complexity of CEO compensation packages. J. Account. Econ. 79 (1), 101709.

Ali, W., Tseng, A., 2025. The Decision Usefulness of ASC 606 Accelerated Revenue Recognition. *Working Paper*.

Amundi, 2023. Voting Report 2023. https://about.amundi.com/files/nuxeo/dl/f03cfbcd-0699-4288-af82-78ad255a95d6.

Armstrong, C., Chau, J., Ittner, C.D., Xiao, J.J., 2025a. Earnings per share targets and CEO incentives. Rev. Account. Stud. 30 (1), 156–201.

Armstrong, C.S., Guay, W.R., Weber, J.P., 2010. The role of information and financial reporting in corporate governance and debt contracting. J. Account. Econ. 50 (2), 179–234.

Armstrong, C.S., Kepler, J.D., Kim, C., Tsui, D., 2025b. Creditor control rights and executive bonus plans. Rev. Account. Stud. 30 (3), 2724–2767.

Atanasov, V., Black, B., 2016. Shock-based causal inference in corporate finance and accounting research. Critical Finance Rev. 5 (2), 207–304.

Autodesk Inc, 2017. Proxy Statement 2017. https://www.sec.gov/Archives/edgar/data/769397/000120677417001531/adsk3175221-def14a.htm.

Banker, R.D., Datar, S.M., 1989. Sensitivity, precision, and linear aggregation of signals for performance evaluation. J. Account. Res. 27 (1), 21–39.

Bebchuk, L.A., Tallarita, R., 2022. The perils and questionable promise of ESG-based compensation. J. Corp. Law 48 (1), 37–75.

Bettis, J.C., Bizjak, J., Coles, J.L., Kalpathy, S., 2018. Performance-vesting provisions in executive compensation. J. Account. Econ. 66 (1), 194–221.

Blankespoor, E., deHaan, D., Marinovic, I., 2020. Disclosure processing costs, investors' information choice, and equity market outcomes: a review. J. Account. Econ. 70 (2), 101344.

Bloomfield, M.J., 2021. Compensation disclosures and strategic commitment: evidence from revenue-based pay. J. Financ. Econ. 141 (2), 620–643.

Bloomfield, M.J., Gipper, B., Kepler, J.D., Tsui, D., 2021. Cost shielding in executive bonus plans. J. Account. Econ. 72 (2), 101428.

Breuer, M., deHaan, E., 2024. Using and interpreting fixed effects models. J. Account. Res. 62 (4), 1183–1226.

Bushman, R.M., Smith, A.J., 2001. Financial accounting information and corporate governance. J. Account. Econ. 32 (1), 237–333.

Cabezon, F., 2025. Executive compensation: the trend toward one-size-fits-all. J. Account. Econ. 79 (1), 101708.

Cadman, B.D., Rusticus, T.O., Sunder, J., 2013. Stock option grant vesting terms: economic and financial reporting determinants. Rev. Account. Stud. 18 (4), 1159–1190.

Carhart, M.M., 1997. On persistence in mutual fund performance. J. Finance 52 (1), 57–82.

Carter, M.E., Lynch, L.J., Tuna, I., 2007. The role of accounting in the design of CEO equity compensation. Account. Rev. 82 (2), 327–357.

Cassar, G., Gibson, B., 2008. Budgets, internal reports, and manager forecast accuracy. Contemp. Account. Res. 25 (3), 707–738.

Chatterjee, C., 2023. Efficiency Gains from Accounting Regulatory Compliance. Working Paper.

Cheng, Q., Cho, Y.J., Yang, H.I., 2018. Financial reporting changes and the internal information environment: evidence from SFAS 142. Rev. Account. Stud. 23 (1), 347–383.

Chenhall, R.H., 2003. Management control systems design within its organizational context: findings from contingency-based research and directions for the future. Account. Org. Soc. 28 (2), 127–168.

Christensen, D., Lynch, D.P., Partridge, C., 2025a. Improvements in investment efficiency prior to a mandated accounting change: evidence from ASC 842. Contemp. Account. Res. 42 (1), 615–648.

Christensen, D., Mittendorf, B., Partridge, C., 2025b. An Examination of Direct and Spillover Effects of Accounting Standards on Firms' Information Environments. *Working Paper*.

Cohen, S., Kadach, I., Ormazabal, G., Reichelstein, S., 2023. Executive compensation tied to ESG performance: international evidence. J. Account. Res. 61 (3), 805–853.

Coles, J.L., Daniel, N.D., Naveen, L., 2006. Managerial incentives and risk-taking. J. Financ. Econ. 79 (2), 431–468.

Cordeiro, J.J., Sarkis, J., 2008. Does explicit contracting effectively link CEO compensation to environmental performance? Bus. Strat. Environ. 17 (5), 304–317.

Core, J.E., Guay, W.R., 1999. The use of equity grants to manage optimal equity incentive levels. J. Account. Econ. 28 (2), 151–184.

Core, J.E., Guay, W.R., 2002. Estimating the value of employee stock option portfolios and their sensitivities to price and volatility. J. Account. Res. 40 (3), 613–630.

Core, J.E., Guay, W.R., Verrecchia, R.E., 2003. Price versus non-price performance measures in optimal CEO compensation contracts. Account. Rev. 78 (4), 957–981.
Core, J.E., Packard, H.A., 2022. Non-price and price performance vesting provisions and CEO incentives. Account. Rev. 97 (7), 109–134.
Dorantes, C.A., Li, C., Peters, G.F., Richardson, V.J., 2013. The effect of enterprise systems implementation on the firm information environment. Contemp. Account. Res. 30 (4), 1427–1461.
Duke Energy Corporation, 2024. Proxy Statement 2023. https://www.sec.gov/ix?doc=/Archives/edgar/data/0001326160/000110465924037589/tm2327443d10-def14a.htm.
Edmans, A., Gabaix, X., Jenter, D., 2017. Chapter 7 - executive Compensation: a Survey of theory and evidence. In: Hermalin, B.E., Weisbach, M.S. (Eds.), The Handbook of the Economics of Corporate Governance. North-Holland, pp. 383–539.
Fama, E.F., French, K.R., 1997. Industry costs of equity. J. Financ. Econ. 43 (2), 153–193.
FASB, 2014. ASU 2014-09 Revenue from Contracts with Customers (Topic 606). https://fasb.org/page/PageContent?pageId=/projects/recentlycompleted/revenue-recognition-summary.html.
FASB, 2016. Leases. https://www.fasb.org/projects/current-projects/leases.
Feltham, G.A., Xie, J., 1994. Performance measure congruity and diversity in multi-task principal/agent relations. Account. Rev. 69 (3), 429–453.
Feng, M., Li, C., McVay, S., 2009. Internal control and management guidance. J. Account. Econ. 48 (2), 190–209.
Field, L., Lowry, M., Mkrtchyan, A., 2013. Are busy boards detrimental? J. Financ. Econ. 109 (1), 63–82.
Flammer, C., Hong, B., Minor, D., 2019. Corporate governance and the rise of integrating corporate social responsibility criteria in executive compensation: effectiveness and implications for firm outcomes. Strateg. Manag. J. 40 (7), 1097–1122.
Gallemore, J., Labro, E., 2015. The importance of the internal information environment for tax avoidance. J. Account. Econ. 60 (1), 149–167.
Gantchev, N., Giannetti, M., Li, R., 2022. Does money talk? Divestitures and corporate environmental and social policies. Rev. Finance 26 (6), 1469–1508.
Gelsomin, E., 2022. Accounting Standards Updates, Investments in Accounting Information Systems, and Firms' Internal Information Environments. Boston College.
Gipper, B., 2021. The economic effects of mandating expanded compensation disclosures. J. Account. Econ. 71 (1), 101338.
Goodman, T.H., Neamtiu, M., Shroff, N., White, H.D., 2014. Management forecast quality and capital investment decisions. Account. Rev. 89 (1), 331–365.
Guay, W.R., 1999. The sensitivity of CEO wealth to equity risk: an analysis of the magnitude and determinants. J. Financ. Econ. 53 (1), 43–71.
Guay, W.R., Kepler, J.D., Tsui, D., 2019. The role of executive cash bonuses in providing individual and team incentives. J. Financ. Econ. 133 (2), 441–471.
Hall, B.J., Liebman, J.B., 1998. Are CEOs really paid like bureaucrats? Q. J. Econ. 113 (3), 653–691.
Heitzman, S., Huang, M., 2019. Internal information quality and the sensitivity of investment to market prices and accounting profits. Contemp. Account. Res. 36 (3), 1699–1723.
Hoberg, G., Phillips, G., 2010. Product market synergies and competition in mergers and acquisitions: a text-based analysis. Rev. Financ. Stud. 23 (10), 3773–3811.
Hoberg, G., Phillips, G., 2016. Text-based network industries and endogenous product differentiation. J. Polit. Econ. 124 (5), 1423–1465.
Hogan, C.E., Wilkins, M.S., 2008. Evidence on the audit risk model: do auditors increase audit fees in the presence of internal control deficiencies? Contemp. Account. Res. 25 (1), 219–242.
Holmström, B., 1979. Moral hazard and observability. Bell J. Econ. 10 (1), 74–91.
Hong, B., Li, Z., Minor, D., 2016. Corporate governance and executive compensation for corporate social responsibility. J. Bus. Ethics 136 (1), 199–213.
Institutional Shareholder Services Inc, 2024. SRI Proxy Voting Guidelines Updates 2024 Policy Recommendations. https://www.issgovernance.com/file/policy/active/specialty/SRI-US-Policy-Updates.pdf.
Jensen, M.C., Murphy, K.J., 1990a. CEO incentives: it's not how much you pay, but how. Harv. Bus. Rev. 68 (3), 138–149.
Jensen, M.C., Murphy, K.J., 1990b. Performance pay and top-management incentives. J. Polit. Econ. 98 (2), 225–264.
Kaplan, R.S., Norton, D.P., 1992. The balanced scorecard: measures that drive performance. Harv. Bus. Rev. 70 (1), 172–180.
Kim, S., Kim, S., Kleymenova, A., Li, R., 2025. Current Expected Credit Losses (CECL) Standard and Banks' Information Production. *Working Paper*.
Laksmana, I., Tietz, W., Yang, Y., 2012. Compensation discussion and analysis (CD&A): readability and management obfuscation. J. Account. Publ. Pol. 31 (2), 185–203.
Li, J., Wu, D., 2020. Do corporate social responsibility engagements lead to real environmental, social, and governance impact? Manag. Sci. 66 (6), 2564–2588.
Linck, J.S., Netter, J.M., Yang, T., 2008. The determinants of board structure. J. Financ. Econ. 87 (2), 308–328.
Mitton, T., 2024. Economic significance in corporate finance. Rev.Corporate Finance Stud. 13 (1), 38–79.
Mummolo, J., Peterson, E., 2018. Improving the interpretation of fixed effects regression results. Polit. Sci. Res. Methods 6 (4), 829–835.
Packard, H.A., Pawliczek, A., Skinner, A.N., 2023. Voluntary performance disclosures in the CD&A. Account. Rev. 98 (6), 435–466.
Pawliczek, A., 2021. Performance-vesting share award outcomes and CEO incentives. Account. Rev. 96 (5), 337–364.
Plumlee, M., Yohn, T.L., 2010. An analysis of the underlying causes attributed to restatements. Account. Horiz. 24 (1), 41–64.
RepRisk, 2023. RepRisk methodology overview. https://www.reprisk.com/news-research/resources/methodology.
Roh, Y., 2025. Internal accounting hiring and operational efficiency: evidence from the implementation of ASC 842. Account. Rev. 100 (5), 317–344.
Shroff, N., 2017. Corporate investment and changes in GAAP. Rev. Account. Stud. 22 (1), 1–63.
Simons, R., 1990. The role of management control systems in creating competitive advantage: new perspectives. Account. Org. Soc. 15 (1), 127–143.
Timmermans, O., 2024. Cash versus share payouts in relative performance plans. Account. Rev. 99 (6), 451–489.
U.S. Securities and Exchange Commission, 2006. Executive compensation and related person disclosure. https://www.sec.gov/rules/final/2006/33-8732a.pdf.